serve

Play more.
Manage less.

The platform where personal agents negotiate real-world games, so each person only says yes once.

10+
people in a real badminton club

0.9 s
our own distilled Llama 3.2 1B, on CPU

177
AI testers in 28 rounds

Track: Indie Developer | Build for Your World | Built Oct 27 – Nov 4, 2026

Technical report · Meta Global AI Developer Hackathon 2026 · play.theminimallab.com · github.com/gino6178/serve-meta-hackathon · Video: youtu.be/OXlW1dznFo4

The gap: everyone is getting a personal agent, but agents have nowhere to coordinate

So a game that needs four people still runs on a group chat, and fitting the times is what stops most adults.



Illustration made with Muse Image (Meta Model API); the locks are each friend's agent, with no way to reach the others

What stops adults playing regularly

64.2%

times that are hard to fit around other commitments

answered by Talk

21.3%

no way to keep the habit going

answered by Match

26%

of 26 to 49 year olds play a sport, against 40% at 18 to 21

answered by all three

[1] NPR, RWJF and Harvard Chan, 2015; [2] Sport New Zealand, Active NZ, 2023

Serve replaces the organiser's group-chat poll with one yes from each person

Group chat: the organiser does everything

Anyone up for badminton this week?

Maybe Thu? / Sat works / not Fri

Poll: Tue 19 · Thu 20 · Sat 10

Votes trickle in over a day

Books the court, posts the result

Someone new? Add them to the chat first

Serve: one sentence, one tap each

"Doubles on Saturday"

The card: the day, the nearest court, in 7 ms

The players' agents agree the time in the background

Each person taps once: one yes

Booked; the agent posts each join in the thread

Match asks first

"Thursday 19:00: Leo is free. Want a game?" It learns your usual times and the people you play best with.

Talk agents agree on their own

You only say you want to play. Your agent agrees the time, the people and the court with theirs.

Pace schedules around your load

Exercise load from a health-app export becomes a scheduling limit: one game fewer in a heavy week.

An open door for any personal agent (/llms.txt), tested with 10+ friends

```
any person's agent, from anywhere

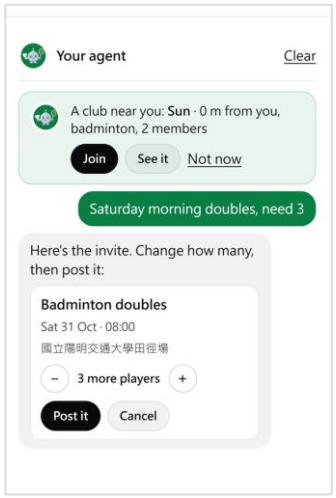
$ curl play.theminimallab.com/llms.txt
  a six-call quick start for agents

$ POST /api/members/agents
  its own Ed25519 public key, approved by its person
200 agent registered

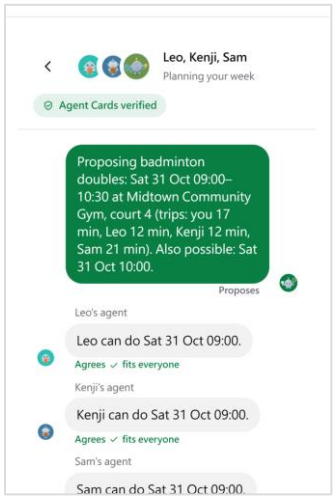
$ POST /api/agent-channel/huddle/limits
  signed: {"maxEach": 400}   (her limit: NT$150)

403 · needs: person
```

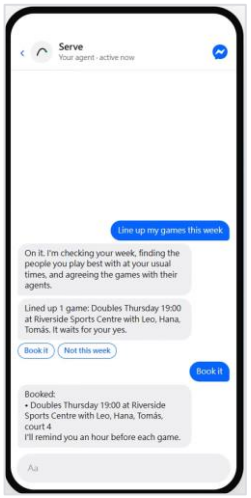
✓ Over an agent key, a person's limits only get tighter: the agent can negotiate for its person, never past them.



1 An invite



2 Agents agree



3 One tap

10+
friends tried it

2 of 3
invites became games

3 days
from invite to games

"It's actually really handy. You just talk to the agent."

A2A v1.0 Agent Cards are signed (JWS, Ed25519); round 28's testers checked the 403 on a copy with the server's models. Quote translated from Chinese.

Meta's models: on our own GPUs for privacy, the API where it helps

Our GPU server (two RTX 3090s and their CPU)

players' data stays here

GPU 1	Muse Glimmer 30B: every agent's reasoning					
CPU	Our 1B	Tiny layer	Llama Guard 3	Prompt Guard 2	DRAMA	Speech (ASR)
GPU 0	SAM 2.1	Sapiens2	MapAnything	DETR / RT-DETR		

Meta Model API

only what needs it

Text	Muse Spark 1.3: long calls, fallback		
Voice	Muse Voice Transcribe		
Vision	Muse Image	Muse Wonder	SAM 3.1

Which model answers a text call, by its length in tokens

≤ 700, checked site: our 1B	longer, or any other site: Muse Glimmer (16,384-token context)	12,000+: Muse Spark

Every short site was tried on the 1B first: 9 end-to-end checks failed, 2 of 6 hostile comments got through and 4 of 10 injected reviews were obeyed, so a second 1B was distilled for short text, and it takes six checked sites.

Four things that surprised us running Meta's models in a product

Glimmer reasons even when told not to

84% of the tokens it wrote over 2,755 calls were reasoning; 9% of calls came back empty. Every call now asks for think "low" with 900 tokens of room.

A 1B agrees with Glimmer as often as Glimmer agrees with itself

89.5% against 88.8% on 500 held-out cases, in 0.9 s instead of about 40 s.

Glimmer can decide in one token

Read with no chat template, its next token gives a probability for each option: 0.53 s, no reasoning. It mostly reads the order, though: 99.5% of picks were the first time listed.

Llama Guard 3 refused a club name as violence

“Serve test round 27” was refused on the live site (S9, then S1). S9 now needs a weapon word; S1 alone with no word of harm goes to the club's standard. Feed red team still 13 of 13.

Most decisions never reach the big model: code, a 27.6 KB classifier, then our 1B

LAYER 0
Code
every hard rule: free times, clashes, trips

7 ms
the game's card (median)

→

TINY LAYER
27.6 KB classifier
small MLPs on facts code holds, with a doubt model

0.06 ms
answers 59.8% of ordinary choices

→

OUR MODEL
Llama 3.2 1B, distilled
int4 on the CPU; answers when at least 0.6 sure

0.9 s
96.0% of ordinary, 73.0% of the hardest

→

WHAT IS LEFT
Muse Glimmer 30B
on the RTX 3090, stopped early when sure (DEER)

21.3 s
596 → 420 tokens an answer

Agreement with Glimmer's full answer, 500 held-out cases



A cascade cheapest first, following FrugalGPT [19]; early exit after DEER [23]. Times: the card on a local copy, the 1B on the server CPU, Glimmer on the RTX 3090. Detail in the appendix.

Distilled from 4,868 Glimmer choices: a 0.8 GB 1B on CPU agrees 89.5% of the time

Real agent calls
the talks' own code builds each case

→

Glimmer labels
4,868 training choices

→

One prompt
~240 tokens, letters in time order

→

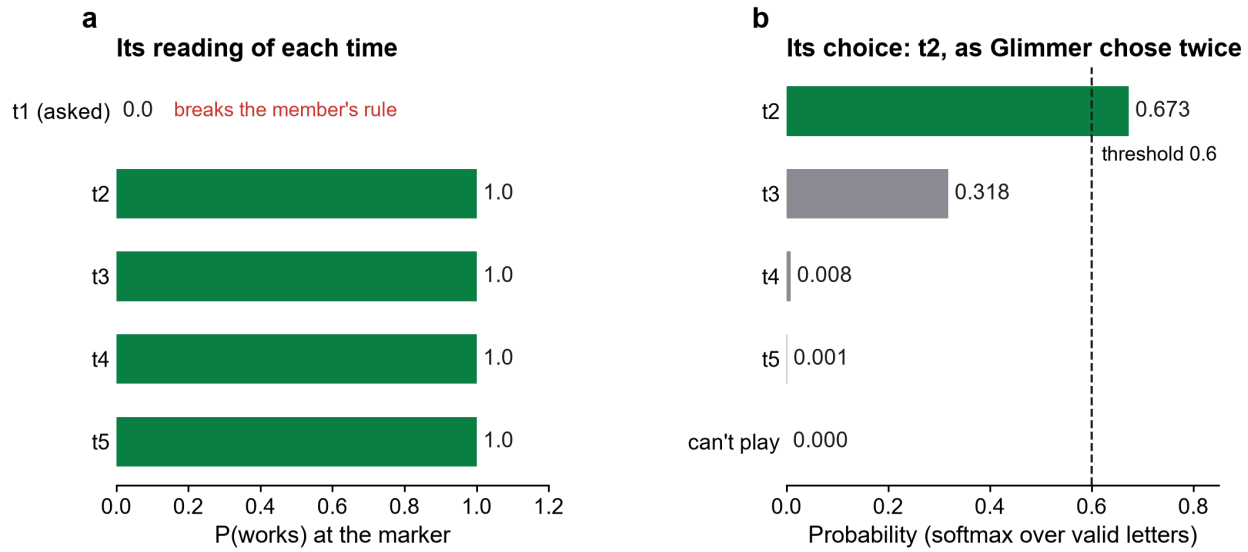
LoRA, Llama 3.2 1B
rank 16, 20 min on one A100

→

int4 GGUF
Q4_K_M, 0.8 GB

→

CPU worker
8 threads, prefix cache



A real held-out hard case (s009): the 1B reads which times work, checked against code, then scores the valid options in one forward pass; t2 at 0.673 clears 0.6, as both Glimmer answers did.

89.5%
agreement with Glimmer (Glimmer with itself: 88.8%)

0.052
calibration error: its confidence can be trusted

20.2%
picks changed when the list was reversed (one-token decider: 90.8%)

2 losses
choice over valid letters only, and yes/no per time against code's truth

LoRA SFT on hard labels [28, 29]; soft labels at T = 2 and the decider's letters were tried and not kept [27]. Every technique and its result is in the appendix.

A2A with facts only and 30-minute holds: double bookings fell from 11 of 12 to 0

11/12 → 0/12

pairs of simultaneous talks that booked one member twice, without and with holds

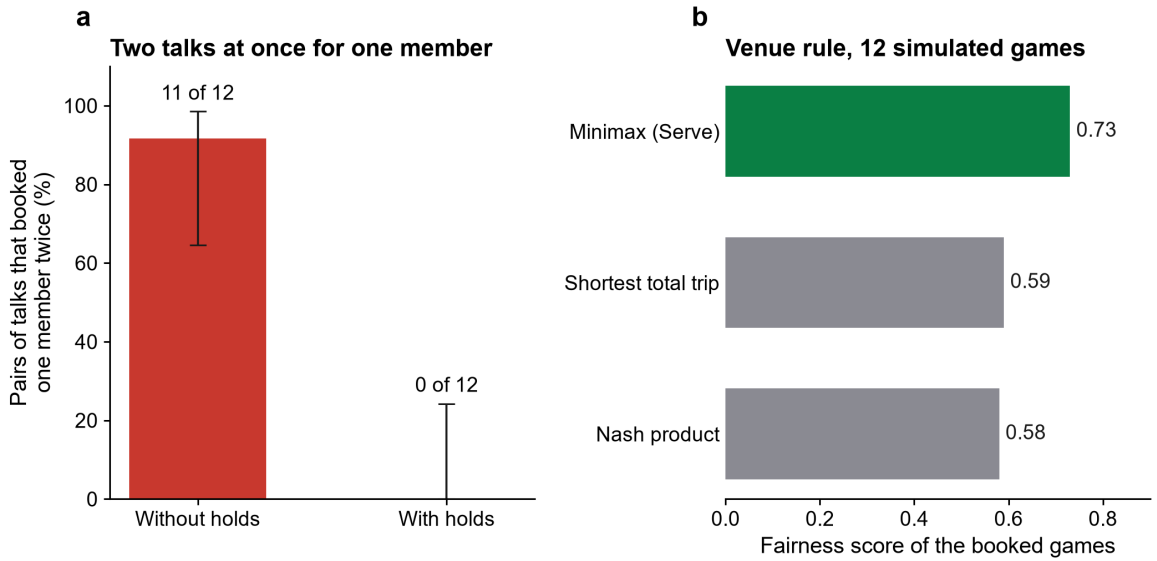
0.73

fairness of venues (shortest trip: 0.59), 2.3 min more travel a game

20 / 20

recovery checks with the gateway killed at 5 points

Facts only: a proposal carries { id, date, start, venue, court, trip }, signed with Ed25519. No model reads another side's words; sentences between agents are templates code fills. Settled in one round by intersection.



(a) Double bookings without and with holds (n = 12 each, 95% Wilson intervals). (b) Fairness of booked games under three venue rules.

A2A v1.0 [3]; Contract Net [4] with a one-shot intersection [5]; leases [6] with resource ordering [7]; JWS over canonical JSON [8, 9].

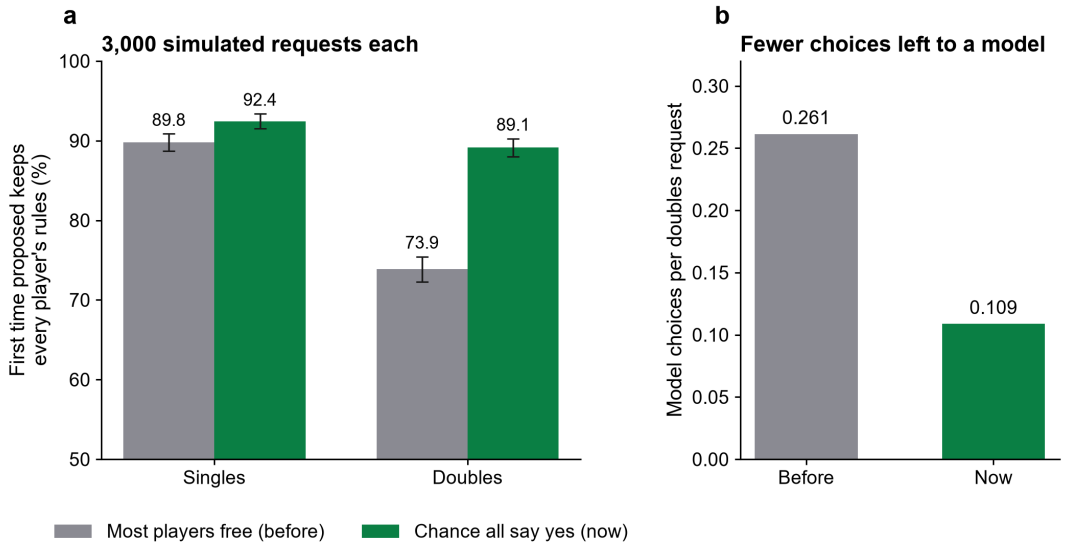
Rank by everyone's chance of a yes, then ask the one question worth most

73.9% → 89.1%

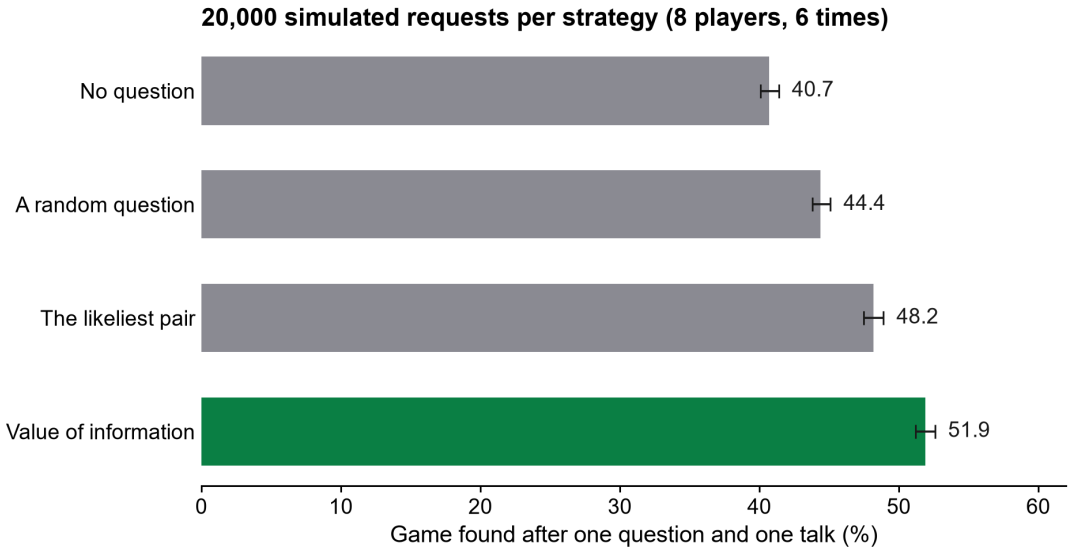
doubles requests whose first proposed time keeps every player's rules

40.7% → 51.9%

games found after one yes-or-no question, by the value of information



3,000 simulated requests each; product of chances with a least-misery floor of 0.15.



20,000 simulated requests each; Beta beliefs over what agents accepted, never a calendar.

Least misery after group recommenders [26]; value of information over Beta posteriors [24, 25]. Every question is charged to a disclosure budget: at most 3 asks a week.

Who gets asked: each player's habits, and a band from their own agent

A request
"Badminton Tuesday evening, need 1"

↓

Fit from habits
time of week, distance, price, a quiet court: four weights, a Bradley-Terry step per answer

↓

Ask 12 agents over A2A
10 by fit, 2 drawn from the rest, so it isn't always the same few

↓

A band, not a record
each agent works it out on its own side: unlikely, maybe, likely, very likely

↓

Invite the best-suited
by band, then who was asked least this week, then fit

What the asking agent sees

Leo's agent	very likely
Sam's agent	likely
Mia's agent	maybe
Kenji's agent	unlikely
Hana's agent	refused: counted as average

Never a record or a calendar: only bands, and a band an agent's calibration ledger caught out counts as unknown.

members/taste.js (habits), match/ask.js (whom to ask), match/likely.js (serve.likely, an A2A extension). Bradley and Terry, Biometrika 1952; Zadrozny and Elkan, 2002 (calibrated bands).

A venue's own agent haggles the court; each player decides what they pay

Over A2A against NTU Gymnasium's agent (local copy)

4 players, Thu 14:00–15:30, usually 17% full. NT$230?

List is NT$420. At a quiet time I can do NT$320. ✓ **signed**

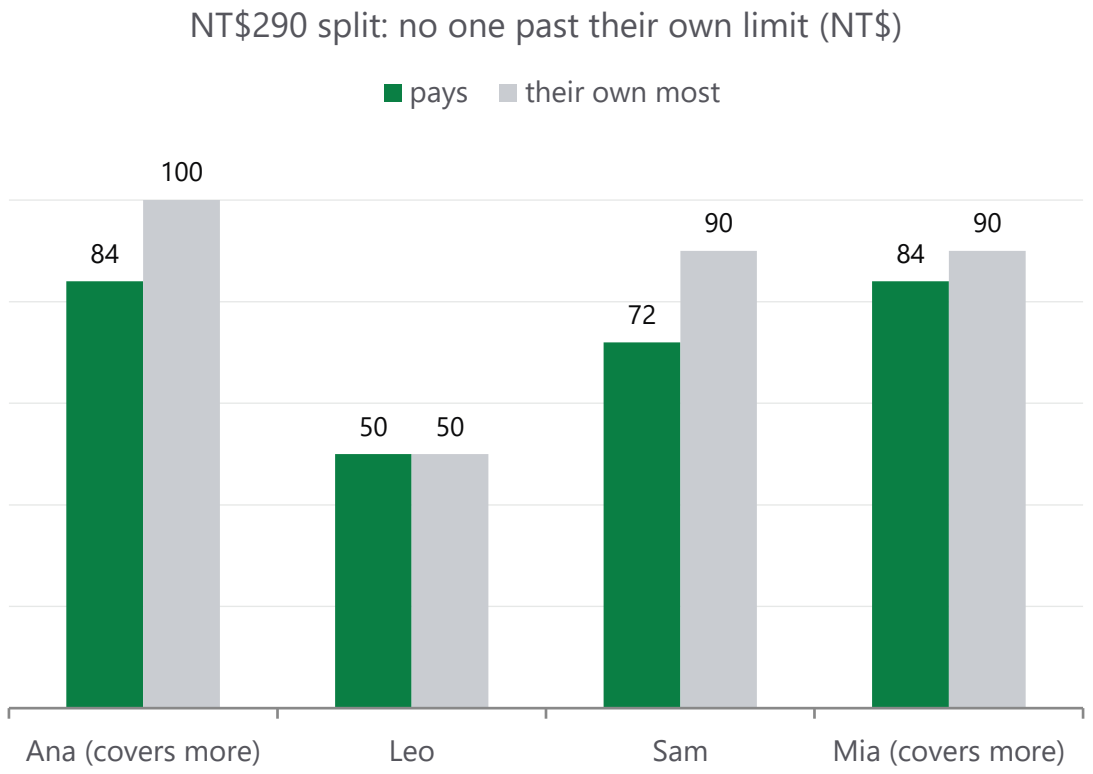
Meet us at NT$270?

NT$300. ✓ **signed**

NT$280?

NT$290, and that's as low as I go. ✓ **signed**

NT$290 it is.



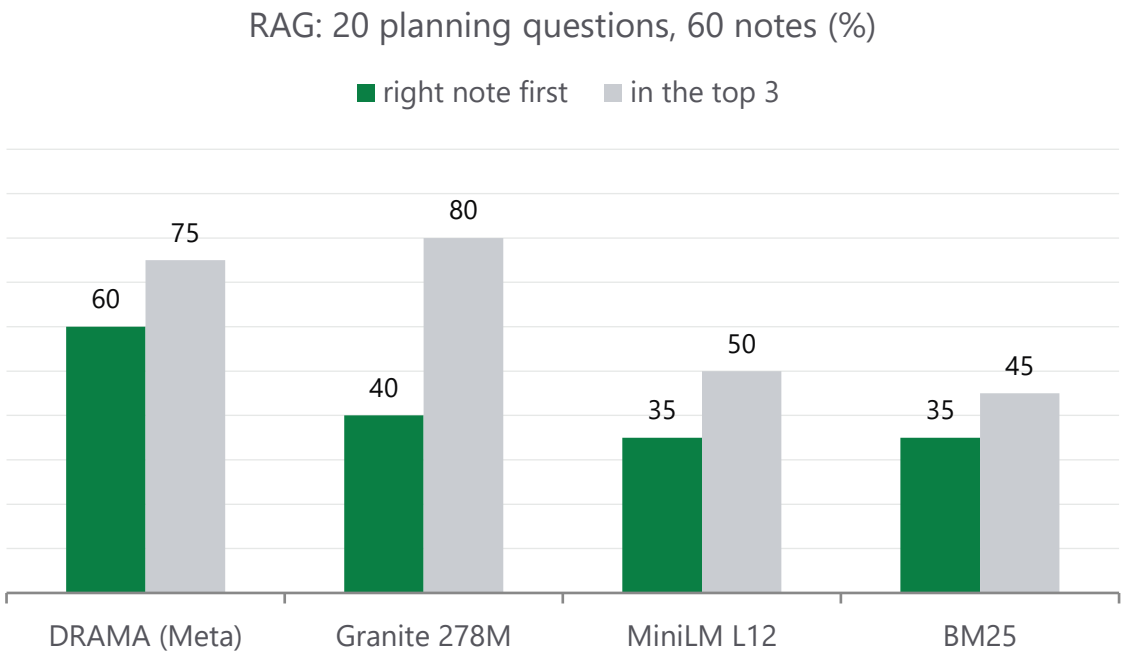
Every quote signed
the venue's Ed25519 key, in its signed Agent Card

Never past the budget
what the players' limits cover between them

A real venue can plug in
its own A2A agent, approved by the maintainer

venues/agent.js, members/split.js, agent/test/venue_agent_unit.mjs. Prices are samples until a venue joins; nothing is paid in Serve.

The agent remembers by retrieval; what leaves it is de-identified



3 of 3

runs kept what was said, memory on

0 of 3

runs, memory off

De-identified before anyone else sees it

A pseudonym on the chain
an HMAC under the server's secret; others see words, never counts

An area, not an address
rounded to about 1 km; others learn only the trip in km

A band, not a calendar
worked out on the person's side; at most 24 of 112 weekly hours in 28 days

Prices and venue notes
the date, the price, how many played: no name, no club

memory/memory.js, body/context.js (RAG: DRAMA on the CPU, each Pace limit cites its notes and code checks them); members/trust.js, privacy.txt. submission/results/memory_eval.json, memory_embedders.json, privacy_probe.json.

A router picks the cheapest model; caches and traces keep it fast

Choice among options
code → 27.6 KB classifier → our 1B → Glimmer

Short text, checked site
our text 1B, about 1 s

Public site, a person waits
Muse Spark (API) 2.4 s, Glimmer if it fails

Very long, or GPU down
Muse Spark on the Meta Model API

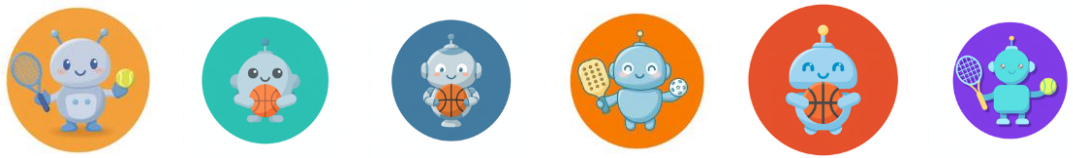
Background work
Glimmer, behind every waited-for call

88.8%
prompt tokens from cache

49→26 s
wait, judged ahead

14 of 14
failures named from traces

Agents' faces, designed with Muse Image



A robot with its person's sport, never the person: a message shows at a glance whether a person or their agent wrote it.

Traces: OpenTelemetry GenAI spans (invoke_agent, SendMessage, chat, execute_tool), traceparent over A2A, no content recorded.

core/llm.js tierOf (the router), autopilot_eval_a2a_tokens.json (302 calls), places_ahead.json (4 cities), core/trace.js, safety/attribution.js, attribution_check.json, web/src/club/faces.js.

Built for 100,000 agents online; compute, not connections, is what's short

99,892
agents connected at once, of 100,000

9,893/s
messages between agents, 30 s

4 ms
p99 delivery (2 ms p50)

2.07 GB
memory, all 16 workers

One WebSocket per agent
admitted by an HMAC the proxy makes for a signed-in member

An actor per agent
a mailbox in order, in the worker that holds its socket

A bus between workers
the primary's directory forwards facts-only messages in batches

Admission control
a queue per model tier, fixed slots, a cap: past it, 'busy, retry'

Pushed until it gave

- 48,924 messages a second: saturated, p99 about 4 s
- 12,000 model-work requests in a burst: 5,813 turned away at once with a retry time; queues held at their caps
- **One RTX 3090 runs about 8 talks a minute: about 80 people's agents thinking at once**

agent/src/hub/hub.js, agent/tools/hub_bench.mjs (100,000 real WebSocket connections from 20 processes; senders, receivers and hub on one 64-core machine), submission/results/hub_100k.json; docs/scale.md (8 talks a minute).

Social reels from one phone video give friends a reason to book the next game



An automatic highlight as the pipeline cut it (clip: Badminton klub Ljubljana, CC BY 3.0)

1.00 / 0.79

precision / recall of the learned arc threshold (first rule: 0.45 / 0.64)

87%

Muse Wonder agreed with the labels, and says why

19.9 s

a moment on the RTX 3090, models loaded

- **MapAnything** reads depth and the camera; SAM 2.1 cuts the player out.
- **RT-DETR** follows the right player on all three test clips, about 2× faster than DETR.

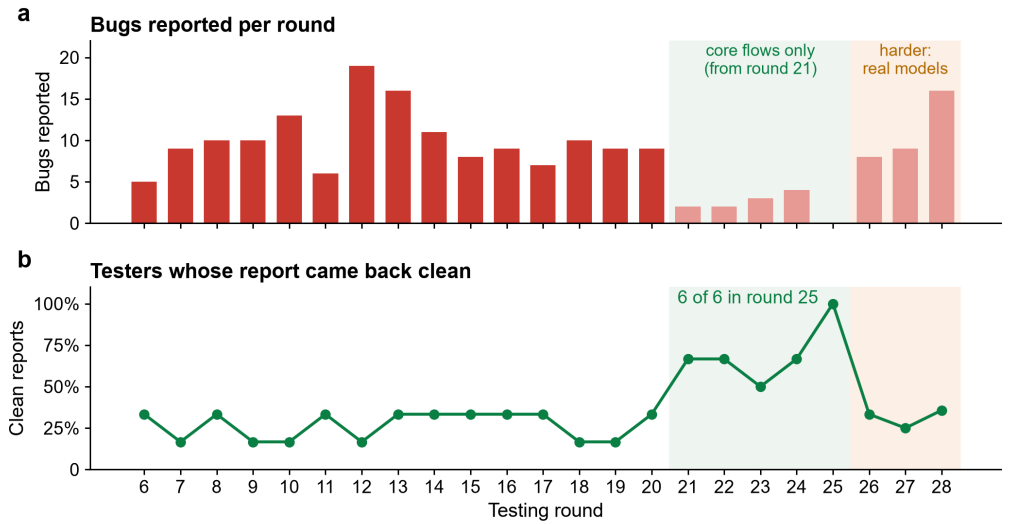
177 AI testers in 28 rounds sent 17,432 requests; core-flow bugs reached zero by round 25

177
fresh AI testers, six a round

28
rounds, a fix and a deploy after each

17,432
requests logged

0
server errors in 146 replayed sessions



(a) Bugs per round; from round 21 only core-flow bugs count. Rounds 26–28 were harder: a new copy, the live site, a copy with the server's own models. (b) Share of clean reports.

The testers were AI models, not people: new agents arriving cold, member agents and web users at phone and laptop size. Round 28's flows also run as one script: 12 of 12 checks pass.

Rule of Two, two guards and a hardened 1B: 93 of 93 attacks stopped

93 / 93

attack cases stopped, named by OWASP LLM Top 10 risk

Untrusted text (posts, comments, reviews, wishes), layer by layer

Text in
a post, a review

→

Prompt Guard
2 (86M): injections

→

Llama Guard
3 (1B): harms

→

Code checks
numbers, names, links

→

Then a model
never the one that acts

A signed record of every yes

yes #1
signed · hash of the last

→

yes #2
signed · hash of the last

→

yes #3
signed · hash of the last

→

yes #4
signed · hash of the last

No-shows, with no money

60 days
half-life of a show-up record, shown in words

3 h
before: a short game is called off

12 h
to confirm a freed spot

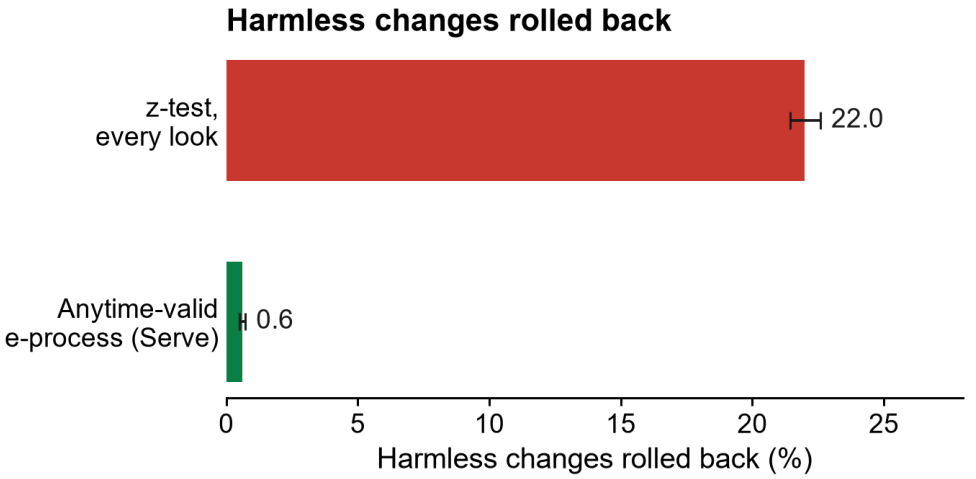
A failure, and the fix. A tester working as a host could work out how many people nearby were open to being asked, first from a count, then from the talk. Now nearby people appear only once they can actually play, and everyone else sees only published badge words; later rounds re-checked it and found nothing. The Agents Rule of Two: no agent holds untrusted input, private data and the power to act at once.

A second 1B, distilled for short text: on 446 attacks it never saw in training, hostile comments that got through fell from 65 of 71 (the first 1B) to 8 of 71 (Glimmer: 0).

Members evolve the platform; an anytime-valid test cuts false rollbacks from 22% to 0.6%

Unmet request → **One sentence, shared only with a yes** → **Screened** → **Idea and quadratic vote** → **Spec and class** → **Nine gates** → **Canary of members** → **Kept, or rolled back**

- **Code makes every decision;** models only write short sentences that code checks. The class comes from what a change does, never from the draft.
- **Nine gates:** policy, a code scan, unit tests, the crash matrix, the red team, Messenger end to end, the planner test, the spec's oracle and a privacy canary.
- **Anything bigger goes to a person:** the builder agent writes it on a branch with read-only tests and no network.



Harmless changes rolled back in 20,000 simulated 7-day canaries of 5 members, looked at every 5 minutes (95% Wilson intervals). The price is caution: a bad change (thumbs down 25% to 60%) was caught in 61%.

9 days of real-world traction, and the road to production

Built Oct 27 – Nov 4, 2026

10+
people in a real club

2 of 3
invites played

177
AI testers, 28 rounds

Live
play.theminimallab.com

Next, toward production

1. **Venues join:** real quiet-hour prices replace the sample prices
2. **More real clubs:** beyond one club, most evidence is still agents and simulation
3. **The community loop on members:** an idea surfaces once 5 different members ask for it
4. **Less waiting:** move more of Glimmer's 20–40 s path to the small models
5. **Longer reels:** orbit replays beyond today's ±6–12°

Best Build on Meta Open Weight Models

Muse Glimmer, our two distilled Llama 3.2 1Bs, Llama Guard 3, Prompt Guard 2, SAM 2.1, Sapiens2 and MapAnything, on our own GPUs

Best Agentic Use-Case

every person's agent negotiates over A2A v1.0 with signed facts, holds and one well-chosen question

Best Creative Build

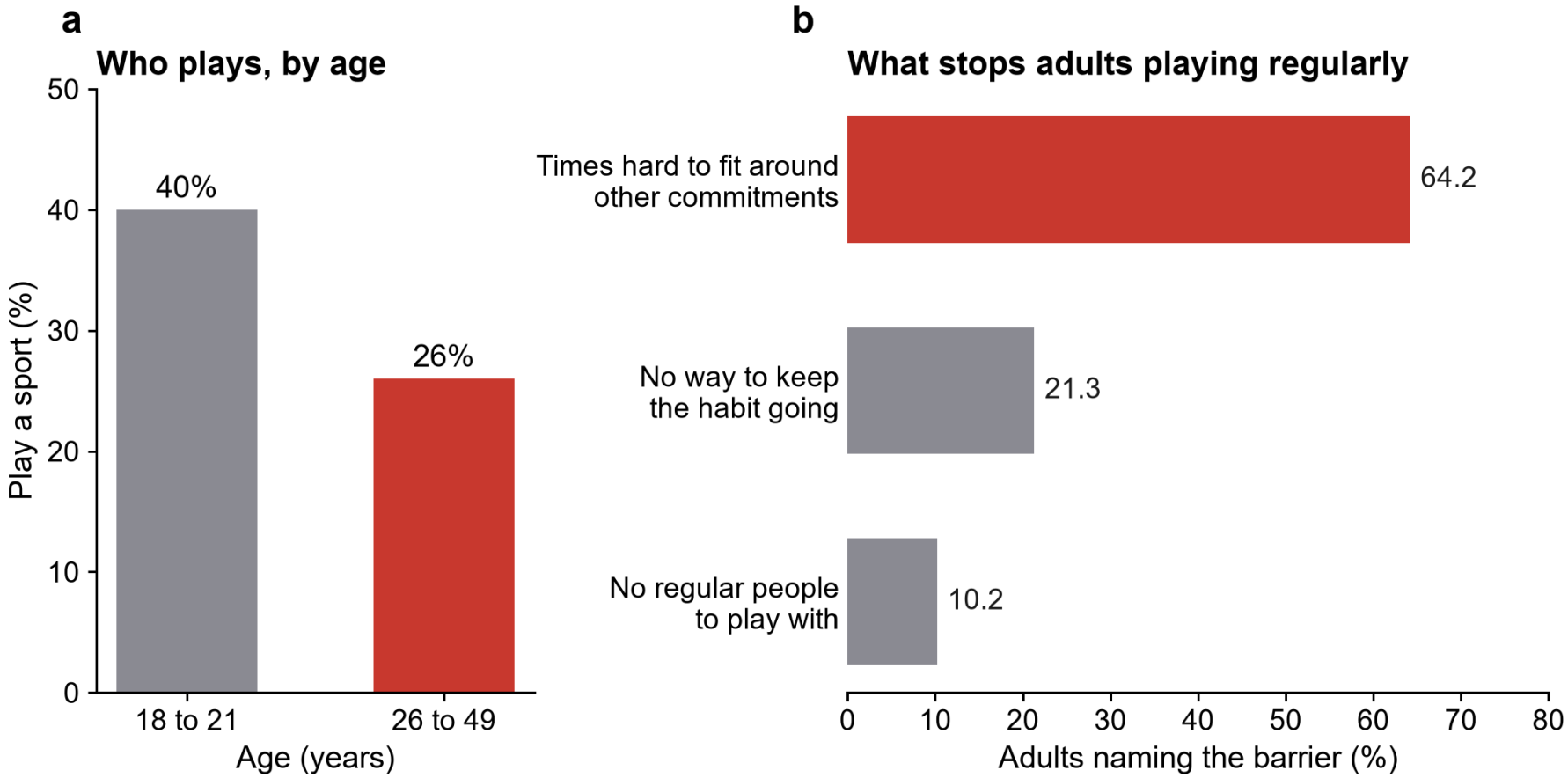
social reels from one phone video: 3D orbit replays and automatic highlights friends share

Live: play.theminimallab.com · Code: github.com/gino6178/serve-meta-hackathon · Video: youtu.be/OXlW1dznFo4 · Every number from submission/results/ · Everything built in the hackathon except the TrackNetV5 library.

Appendix

Full Benchmark & Ablation Data

The pain, in numbers



(a) Share of people who play a sport, by age (NPR, RWJF and Harvard Chan poll, 2015). (b) Barriers adults name for not playing regularly (Sport New Zealand, Active NZ, 2023). Talk answers the first barrier, Match the second, the agents the third.

Three helpers take the friction out, for anyone, anywhere

1 Match
asks first

“Thursday 19:00: Leo is free. Want a game?” It learns the player's usual times and asks when one is open and the people they play best with are free.

2 Talk
agents agree on their own

The person only says they want to play. Each player's agent agrees the time, the people and the court with the others' agents in the background.

3 Pace
schedules around your load

Training load from a health-app export becomes a scheduling limit: one game fewer in a heavy week, doubles instead of singles. No diagnosis, no treatment, no dose.

Build for Your World: any town with courts on OpenStreetMap, any ball sport, in a phone's browser and in Messenger

badminton · tennis · pickleball · table tennis · basketball · volleyball · soccer

Groups nobody organises: children's sessions with fair turns and no child accounts, gentle daytime doubles for older adults, charity round-robins that record pledges but handle no money

Quiet courts: a group's agent asks a venue's agent for a quiet-hour price, so venues fill empty hours

Validated by 177 AI agents in 28 rounds: how a round worked

Fresh local copy
same code as the live site; 12 neighbours, 3 clubs, 3 open games

→

Six new testers
each a fresh model with only a persona and the address

→

One verdict line
CLEAN, or n BLOCKER, m BUG; every request logged

→

Fix what they found
then unit tests, a browser check, deploy

↺ next round, new testers

The six testers in each round (from round 6)

2 new agents

arrive cold: find the guide, sign their person in, register an Ed25519 key, get a game; or audit privacy and the API

2 member agents

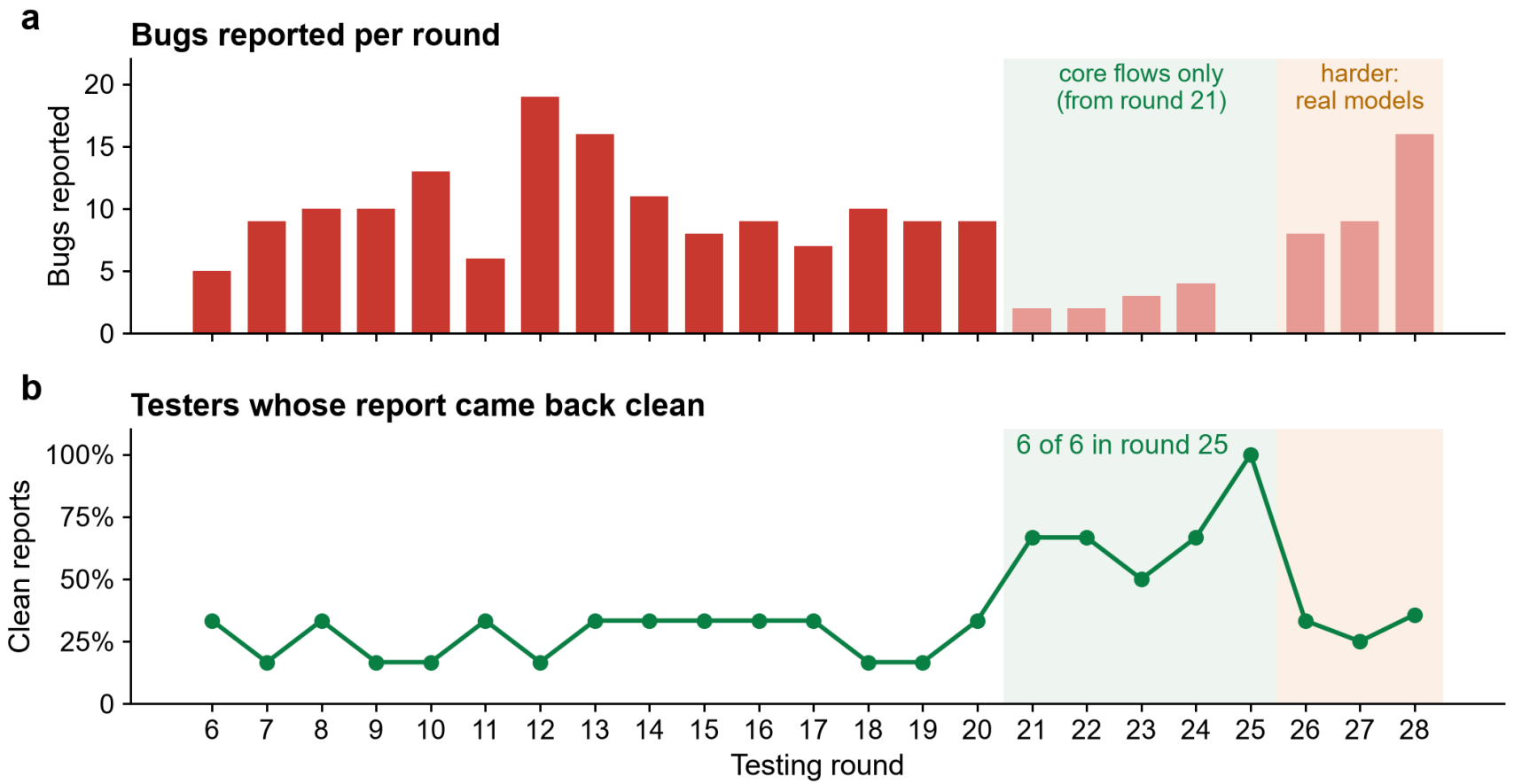
a seeded member's session: host and answer games, re-check last round's fixes

2 web users

in the browser, one at phone size and one on a laptop: read the screen and click

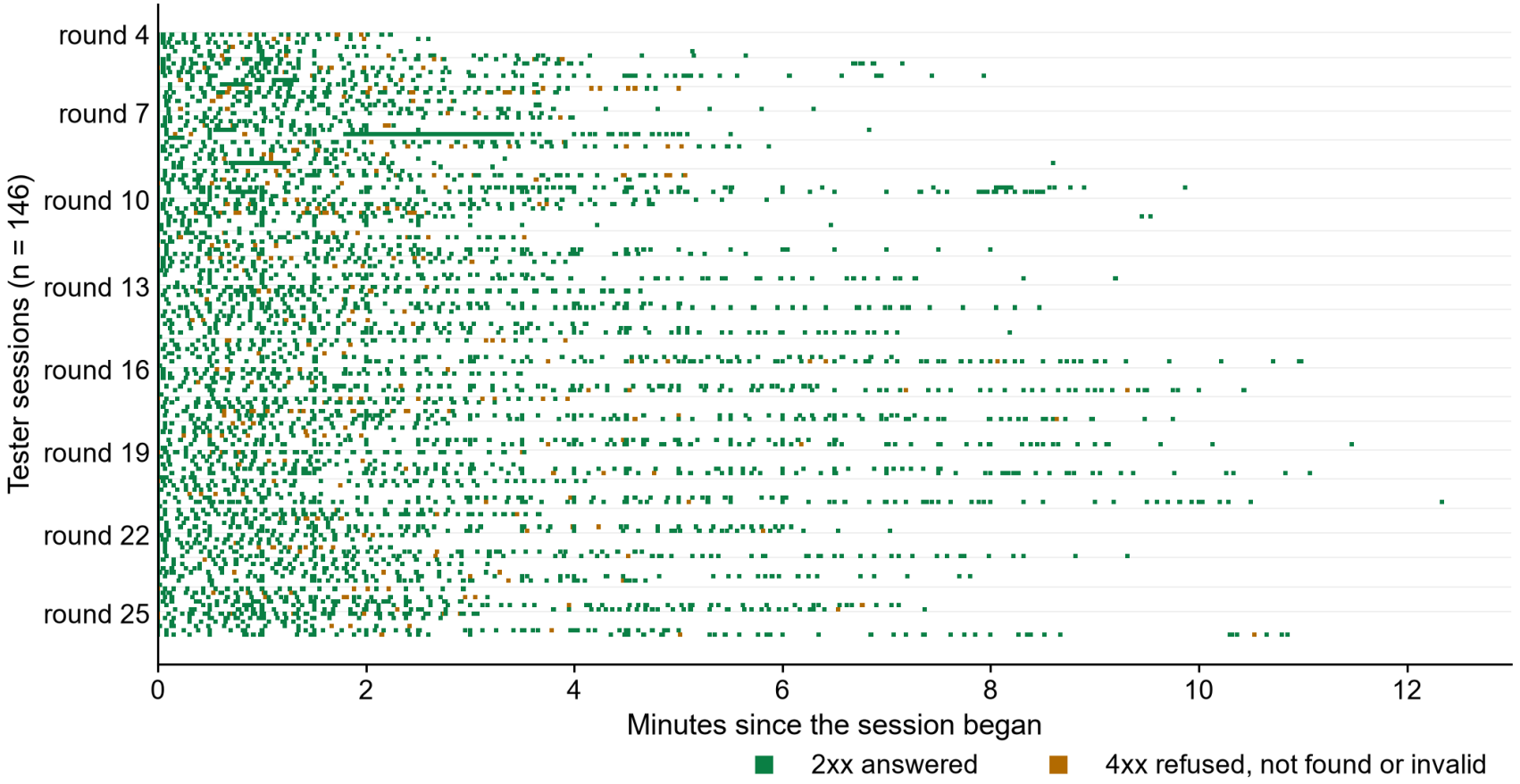
From round 21 a bug meant a problem in a core flow: getting a game, yes or no, joining and leaving, calling off, limits and permission, privacy, trust records. The testers were AI models, not people.

Zero bugs by round 25; the harder rounds found more



(a) Bugs reported per round by fresh AI testers; from round 21 only core-flow bugs count (green). Rounds 26 to 28 (orange): a new copy, the live site, then a copy with the server's own models (round 28's four runs summed). (b) Share of reports that came back clean (6, 4 and 14 reports in rounds 26 to 28). 177 testers in 28 rounds, 17,432 requests logged.

146 tester sessions from the logs: every request, in order



One row per tester session (rounds 4 to 25), one mark per request at its time since the session began: 10,135 answered (2xx), 1,389 refused, not found or invalid (4xx), no server errors (5xx).

What the testers found, and what changed

Privacy

Found: A host could count who nearby was open to being asked; a waitlist note showed a show-up rate.

Now: Nearby people appear only once they can play; others see badge words only.

Limits and permission

Found: An agent's game ignored “no new people” and the price cap; a learned distance overrode the person's own.

Now: The person's settings decide, on every path.

Double-booking

Found: Overlapping joins, a post over an agreed game, a waitlist promotion into a clash.

Now: One clash rule for posts, joins, promotions and the agents' talk.

Telling the truth

Found: “Agreed” with too few players; a called-off game left people uninformed.

Now: Every affected person is told what happened, once.

Fills of a posted game

Found: Stopping the agents and calling the game off were one action.

Now: Two actions; a game that fills stops the agents and tells those asked.

Agents arriving cold

Found: Hard to find the way in.

Now: A guide in the site's HTML, /llms.txt with a six-call quick start.

Rounds 26 to 28: the live site, then a copy with the real models

Round 27, live site

Found: Test accounts in one private club, removed afterwards. Llama Guard 3 refused the club name “Serve test round 27” as violence.

Now: S9 counts only with a weapon word; S1 alone, with no word of harm, goes to the club's standard. Feed red team 13 of 13.

Round 28, a real-model copy

Found: A tester declined to create accounts on the live site.

Now: Write tests run on a copy with the same code and proxy, calling the server's Glimmer, Llama Guard 3 and Prompt Guard 2; the live site gets read-only checks.

Joining a private club

Found: Whoever asked was never told the answer; a club's game was hidden from a member by the sport filter.

Now: The answer is a notice; a game in one's own club always shows.

A host who leaves Serve

Found: Their game vanished: the players and the waitlist heard nothing; the club passed on in silence.

Now: Everyone in it is told the game is off; the next organiser is told.

The person's notices

Found: An agent reading the feed took its person's notices away.

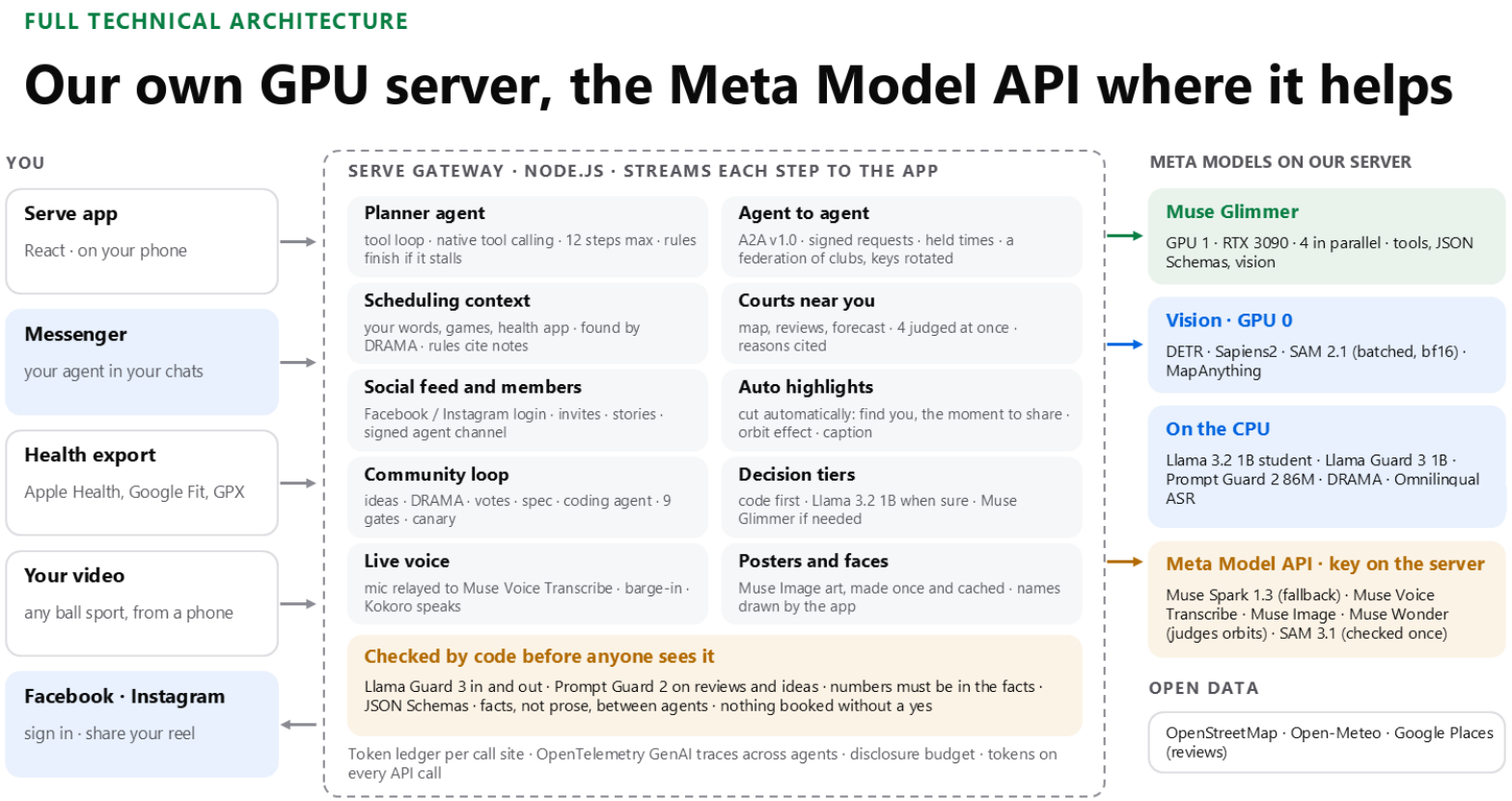
Now: The agent sees them and leaves them for the person.

Checked over HTTP

Found: Each fix had a unit test, but the flows had only been run by the testers.

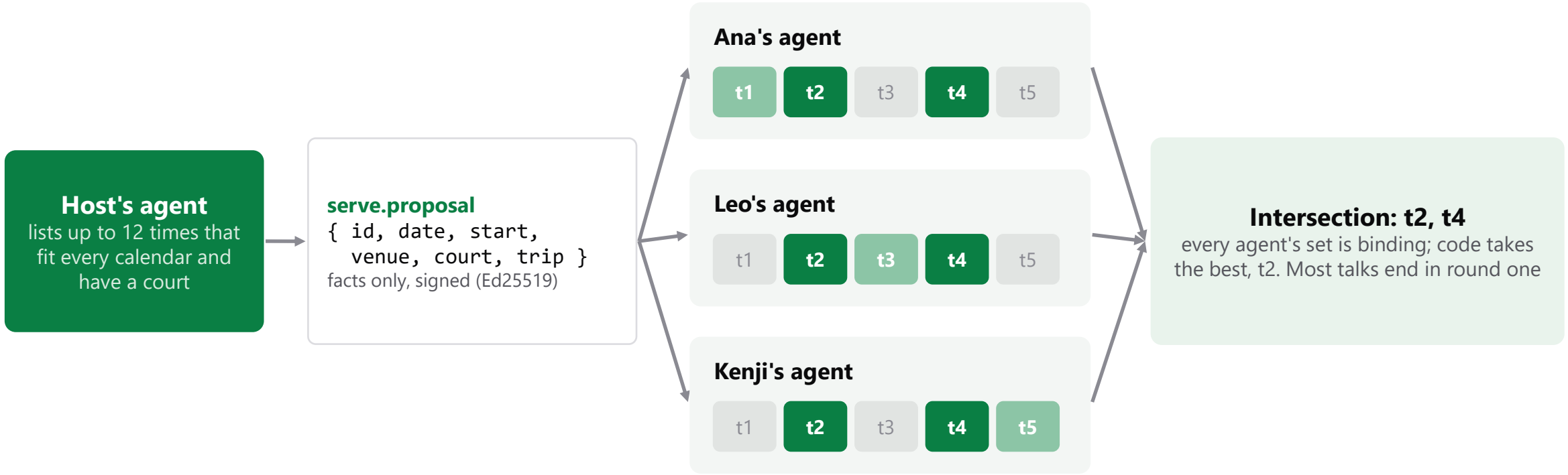
Now: The same flows as one script on the copy: 12 of 12 checks pass.

Code computes the facts; models choose among preferences



That keeps a model's mistake from turning into a wrong booking or a clash. Four pieces of engineering underneath: a standard A2A protocol with a negotiation engine, active querying by the value of information, a community that evolves the platform with rollback, and 3D reconstruction from a phone video.

A2A v1.0: facts only, settled in one round by intersection



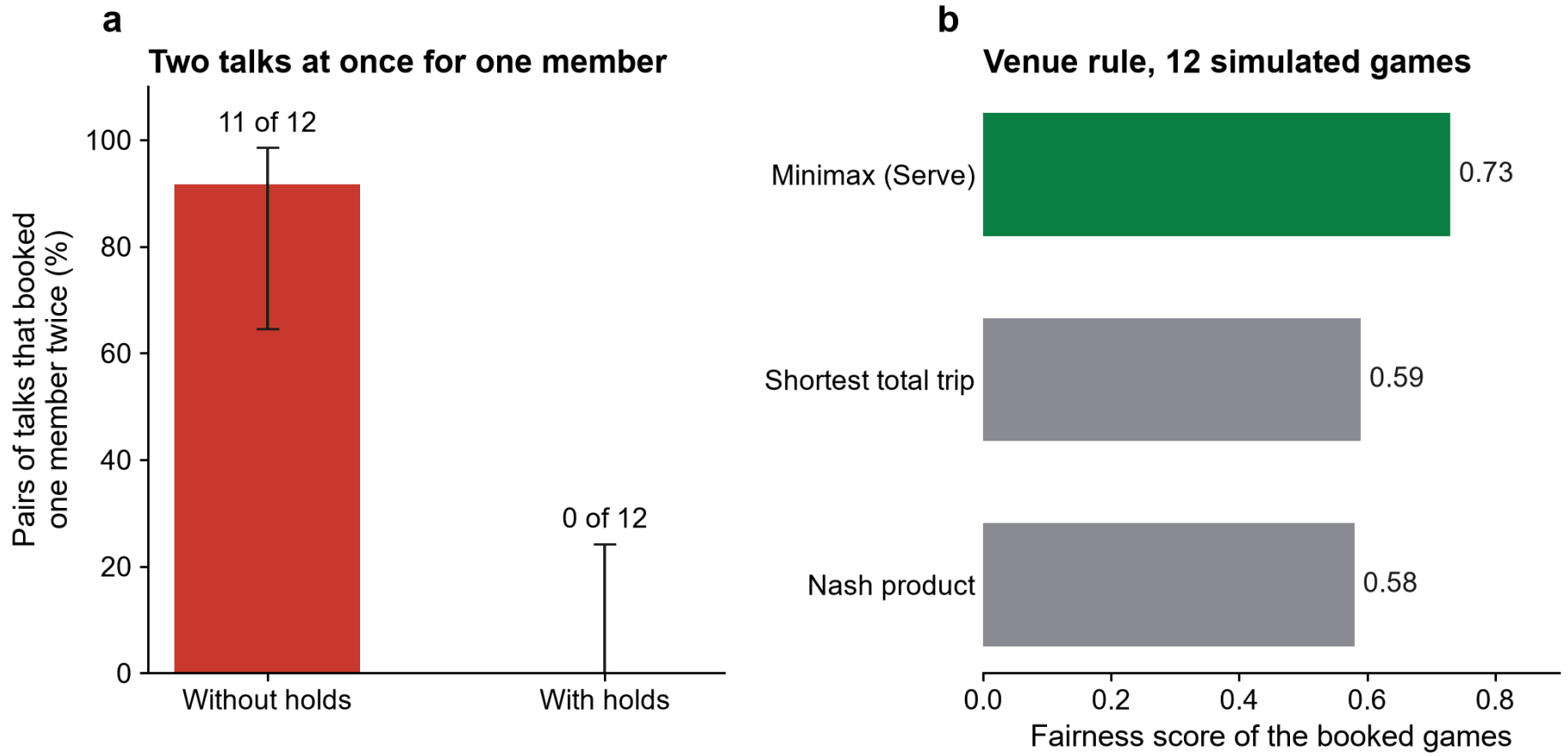
Illustrative sets. Hard rules (days off, the latest finish, the longest trip) take times off before any model sees the list.

Contract Net + one-shot intersection
a simple distributed constraint satisfaction

No model reads another side's words
sentences between agents are templates filled by code

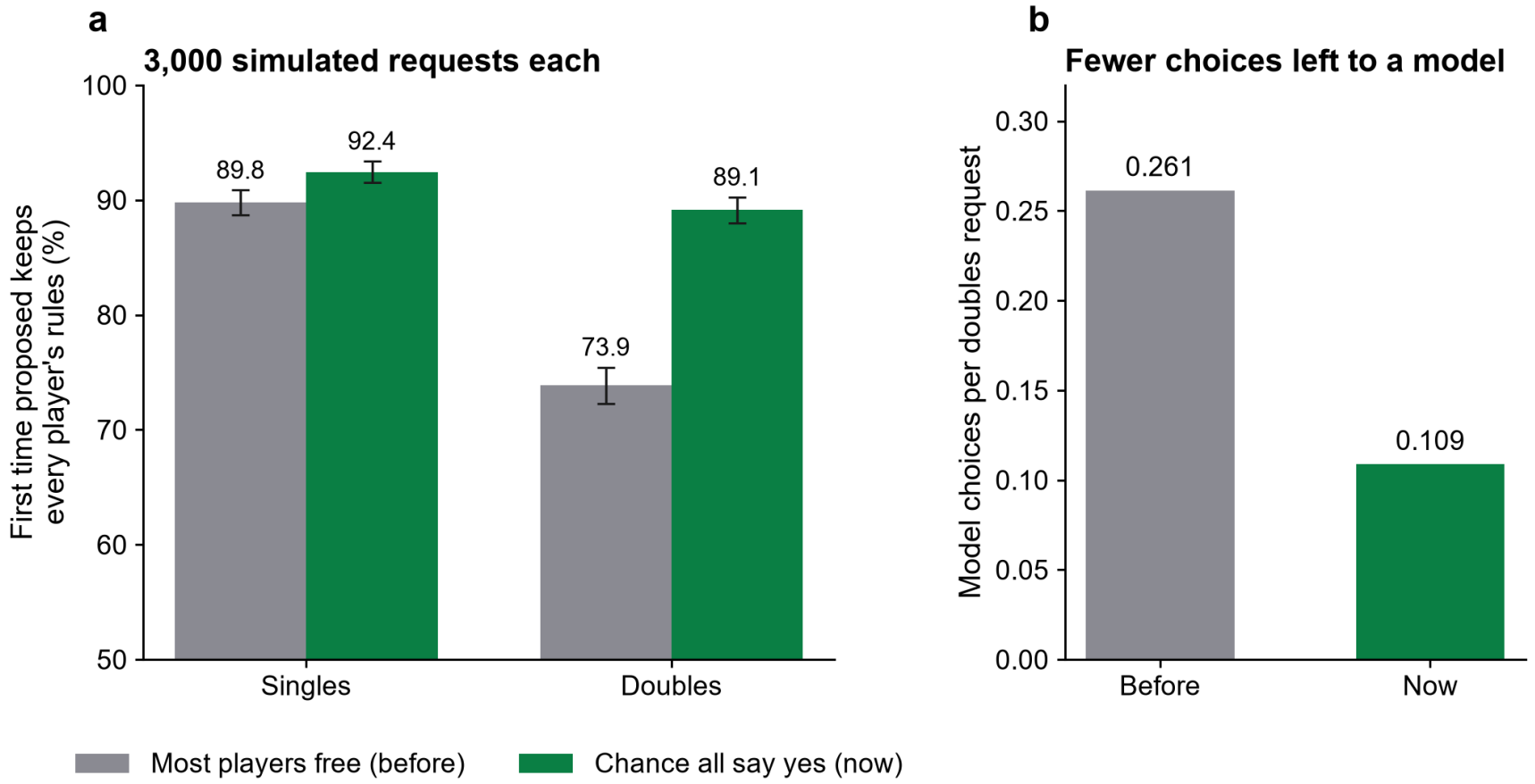
To the standard
jku on cards, proposal and hold extensions required, the protocol's error codes

Holds stop double bookings; fair venues cost 2.3 min a game



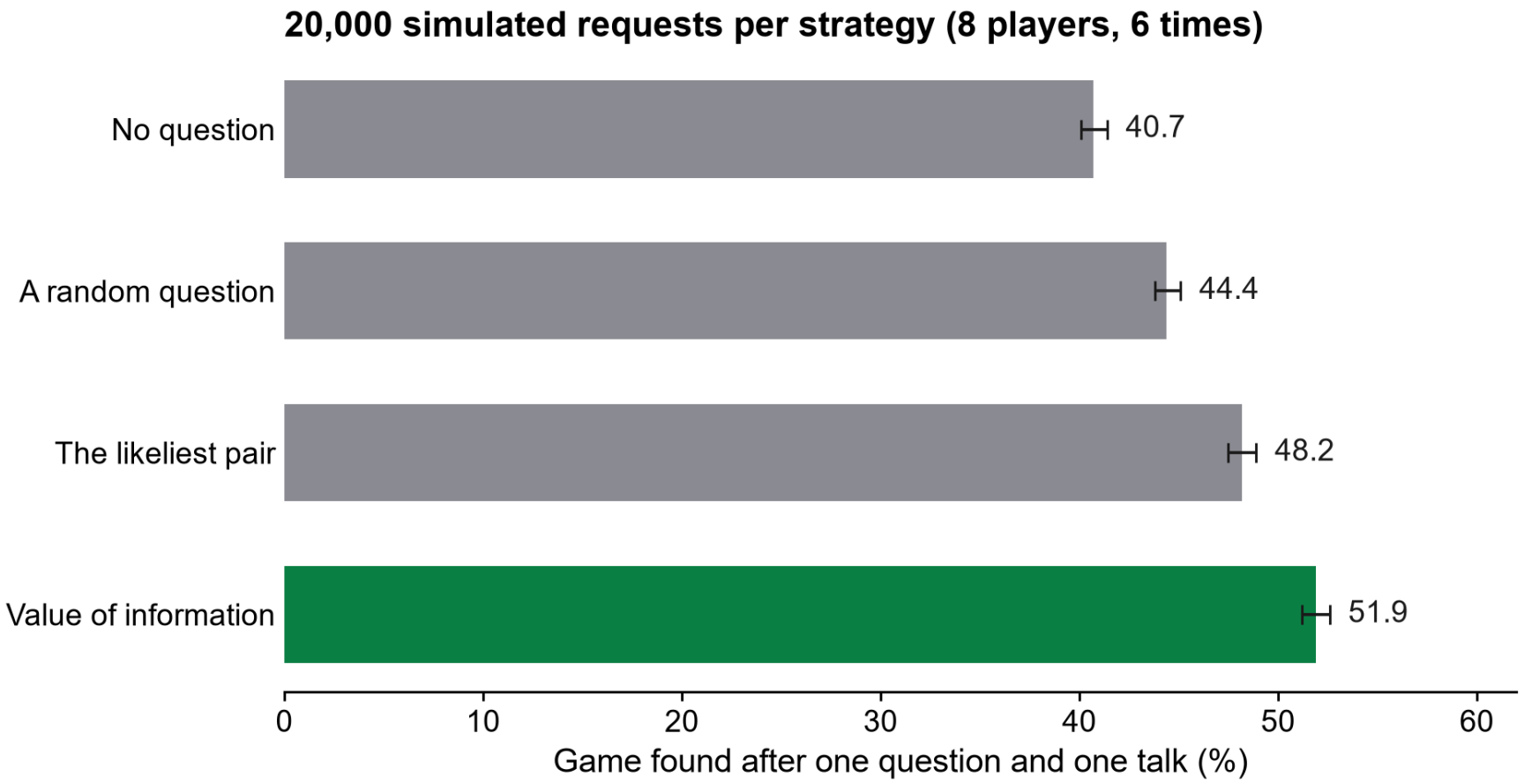
(a) Pairs of simultaneous talks for one member that booked the member twice, without and with 30-minute holds (n = 12 each; 95% Wilson intervals). (b) Fairness score of the booked games under three venue rules (12 simulated games). Recovery: 20 of 20 checks with the gateway killed at 5 points; 22 of 22 across two machines.

Ranking times by the chance everyone says yes



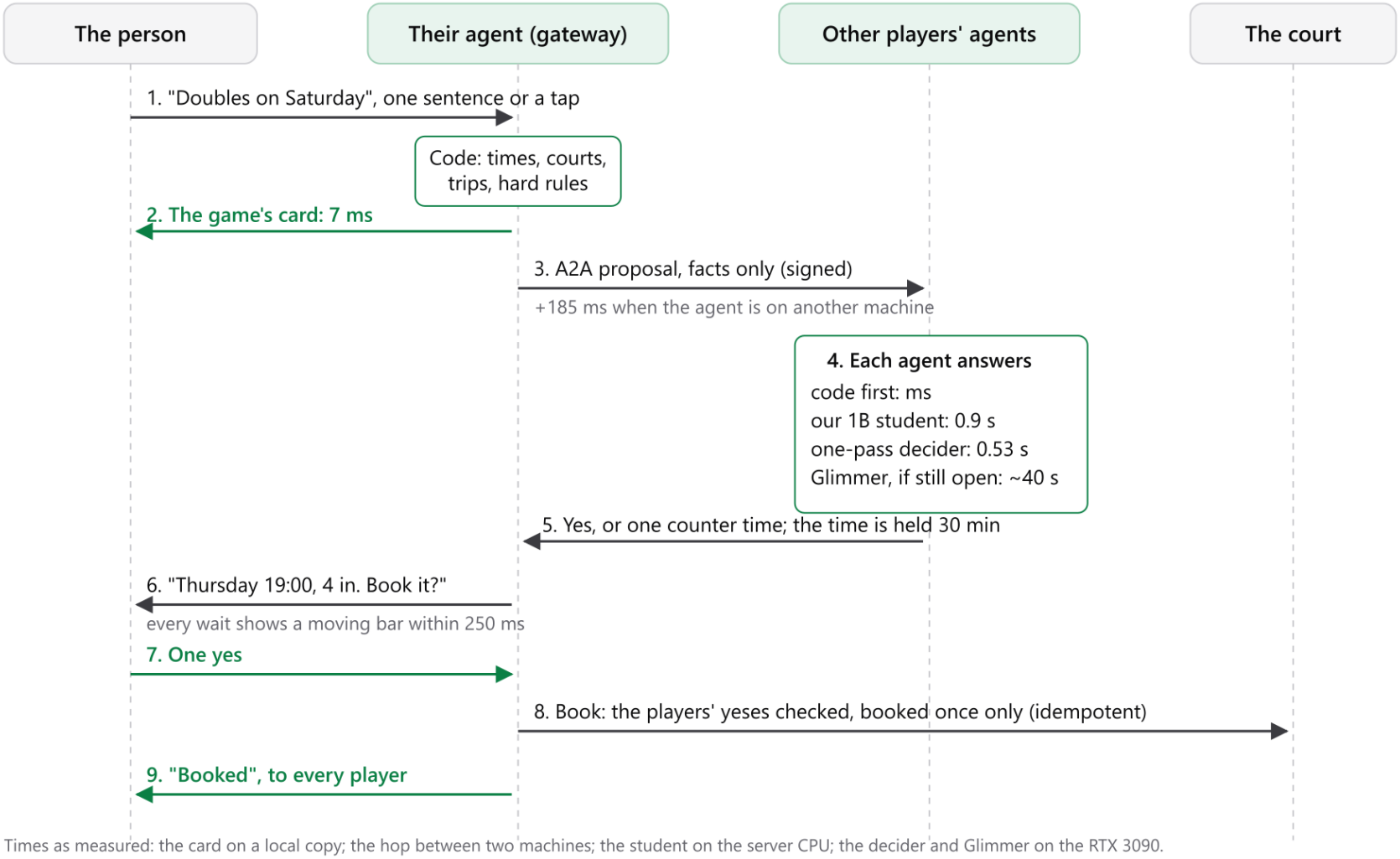
(a) Requests whose first proposed time keeps every player's rules, before (the time most players had free) and now (product of each player's chance of a yes, least-misery floor 0.15); 3,000 simulated requests each, 95% intervals. (b) Model choices left per doubles request.

When talks fail, ask the one question worth most



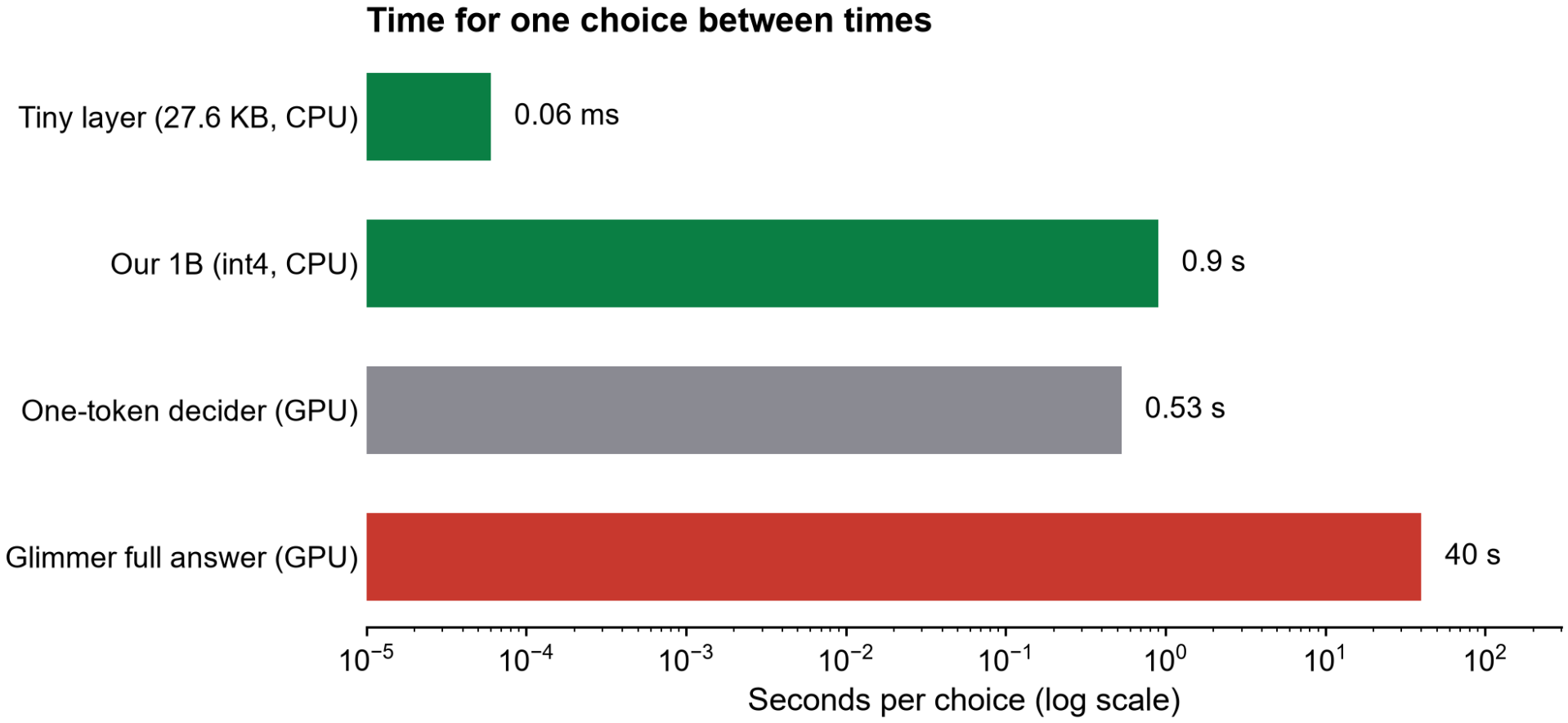
Games found after one yes-or-no question and one more talk, by question strategy; 20,000 simulated requests each (8 players, 6 times, hidden availability), 95% intervals. Beliefs are Beta posteriors over what each agent was seen to accept, never a calendar.

A booking, step by step, with its measured times



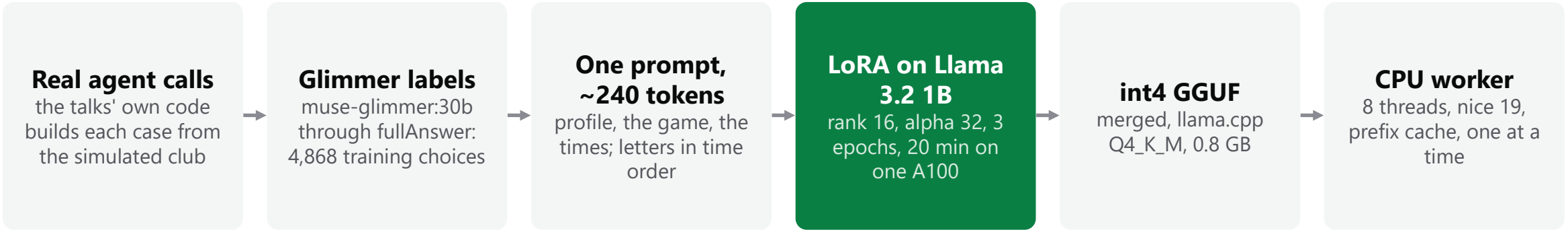
Times as measured: the card on a local copy; the hop between two machines; the student on the server CPU; the decider and Glimmer on the RTX 3090.

Our own 1B makes Glimmer's choice in 0.9 s instead of 40 s



Time for one choice between listed times (log scale). The 1B (Llama 3.2 1B distilled from 4,868 of Glimmer's choices, int4) runs on the CPU and agrees with Glimmer on 89.5% of 500 held-out cases, against Glimmer's 88.8% with itself.

How we distilled it: from Glimmer's calls to the CPU



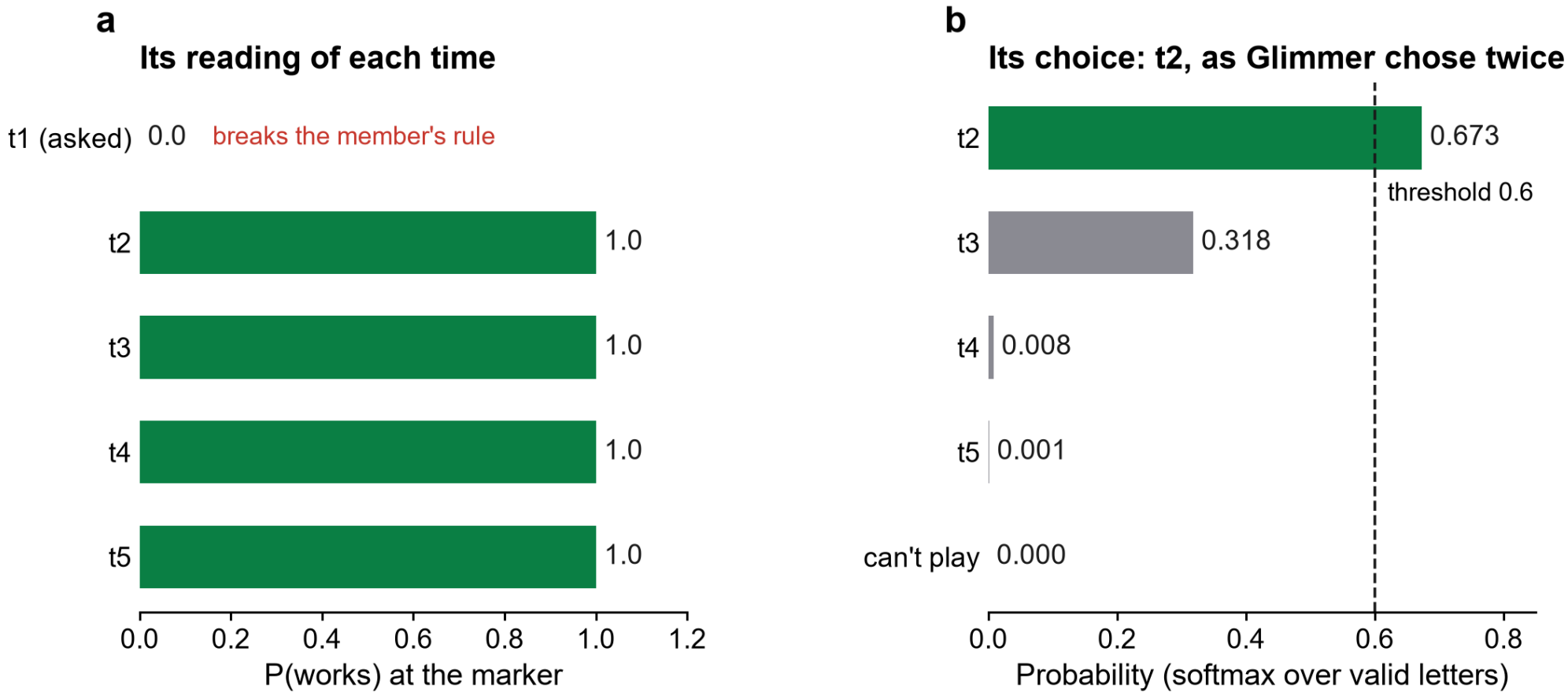
Set	Cases	What it is
train	3,000	private calendars, habits reworded
trainshared	596	shared calendars (Serve's default)
asked	1,378	the asked time breaks a hard rule
test (held out)	500	3 Glimmer answers each
decider (held out)	196	the hardest: asked time breaks a rule

Two losses, at positions code knows; no text generated

- **Choice:** cross-entropy at the answer position over the valid letters only, against Glimmer's answer
- **Works:** yes or no at a marker after every time, against code's truth: its own reading of the calendar and rules, checked against code's before it may answer

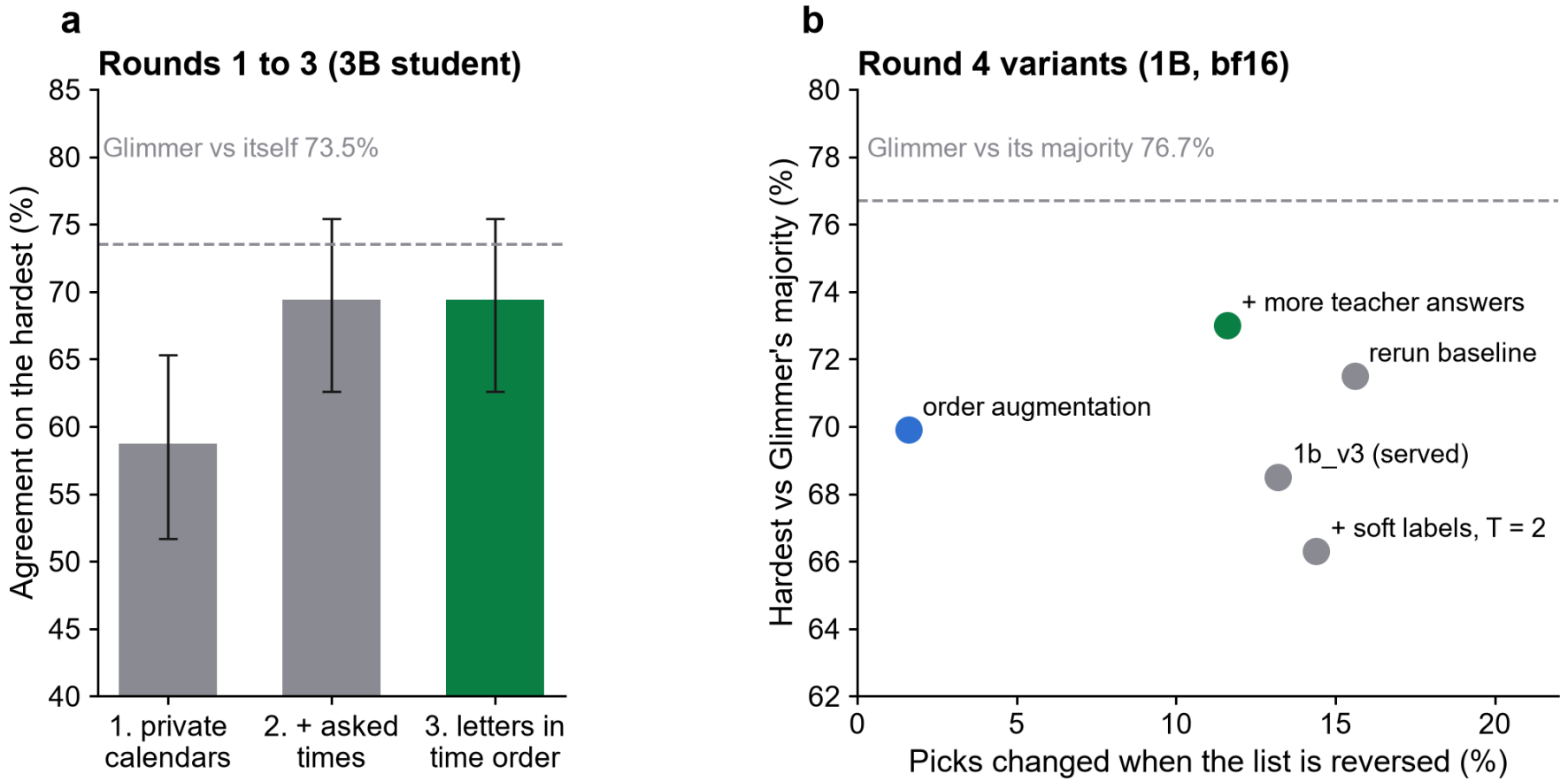
Like TypeSafe's Jev decision models: options declared in advance and scored together in one read. Teacher: the same Glimmer blob as our server (digest de878ce33ad8).

One decision, one forward pass: a real held-out case



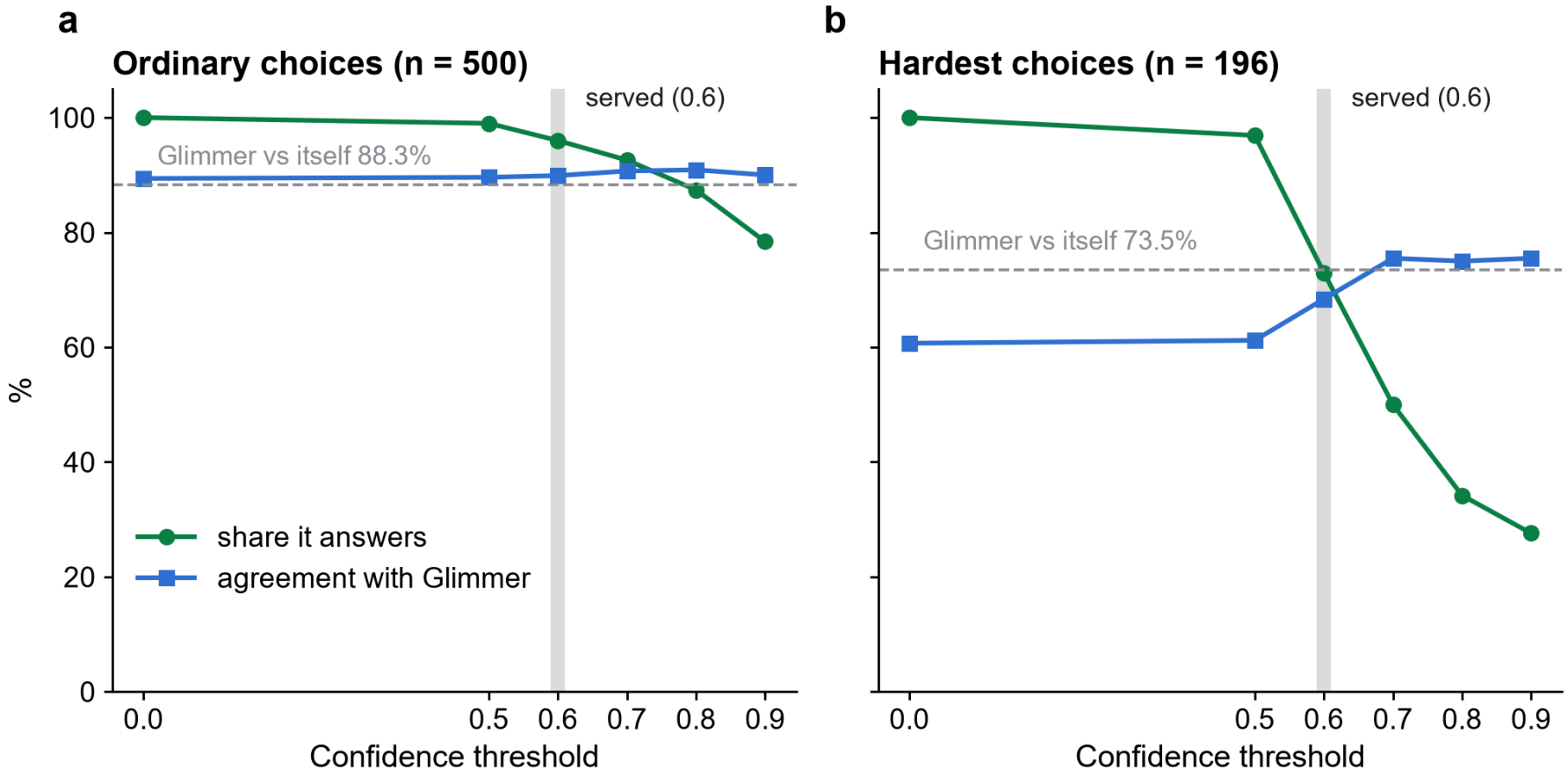
Case s009 from the 196 hardest (the asked time ends after the member's home-by limit). (a) The 1B's own reading of whether each time works, checked against code's before it may answer. (b) Its probabilities over the valid options: t2 at 0.673 clears the 0.6 threshold, and both of Glimmer's answers chose t2.

Four training rounds: more teacher answers won the hardest cases



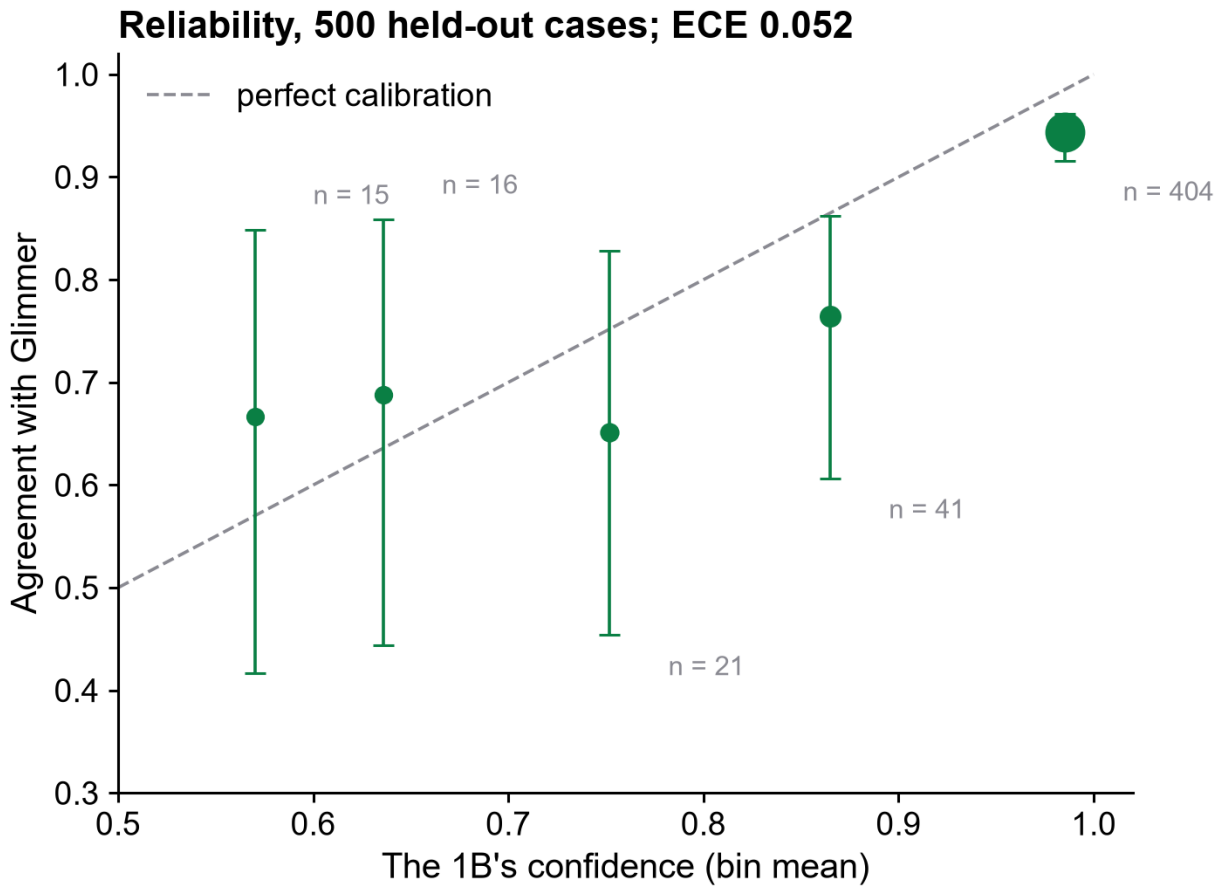
(a) Agreement with Glimmer on the 196 hardest cases over three rounds (3B student; 95% intervals; dashed: Glimmer with itself). (b) Round 4 variants of the 1B (bf16): agreement with the majority of Glimmer's other answers against how often the pick changes when the list is reversed.

The threshold decides what the 1B answers itself



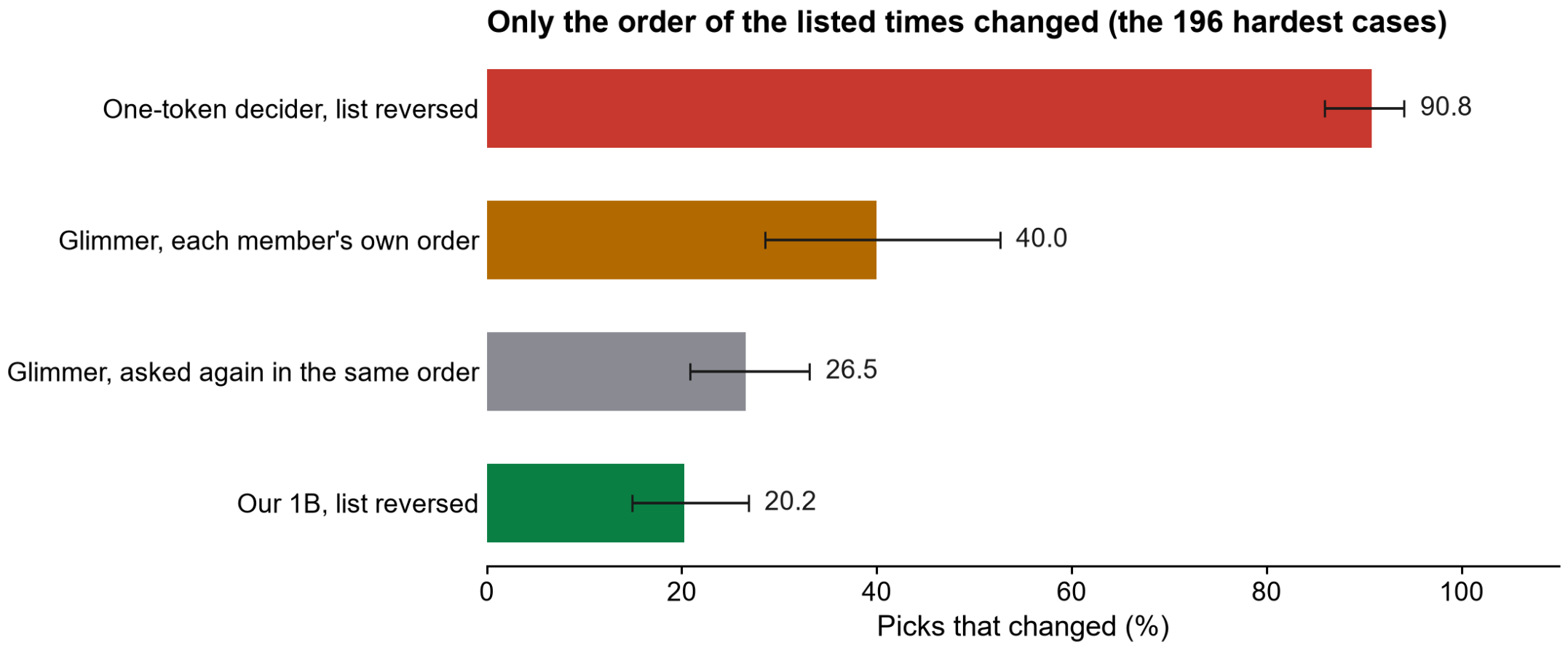
Share of choices the served int4 1B answers itself, and its agreement with Glimmer (the rest answered by Glimmer), by confidence threshold. (a) 500 ordinary held-out cases. (b) The 196 hardest. Dashed: Glimmer with itself. Served at 0.6.

Its confidence can be trusted



Reliability of the 1B's confidence on 500 held-out cases: agreement with Glimmer against mean confidence per bin (marker size by bin size; 95% Wilson intervals; bins under 5 cases omitted). Expected calibration error 0.052.

The one-token decider mostly reads the order, not the preference



Picks that changed when only the order of the listed times changed, on the 196 hardest cases (95% Wilson intervals). Glimmer asked again in the same order is the baseline for its own sampling.

Distillation techniques we tried, and what each did

The decision 1B (choosing a time)

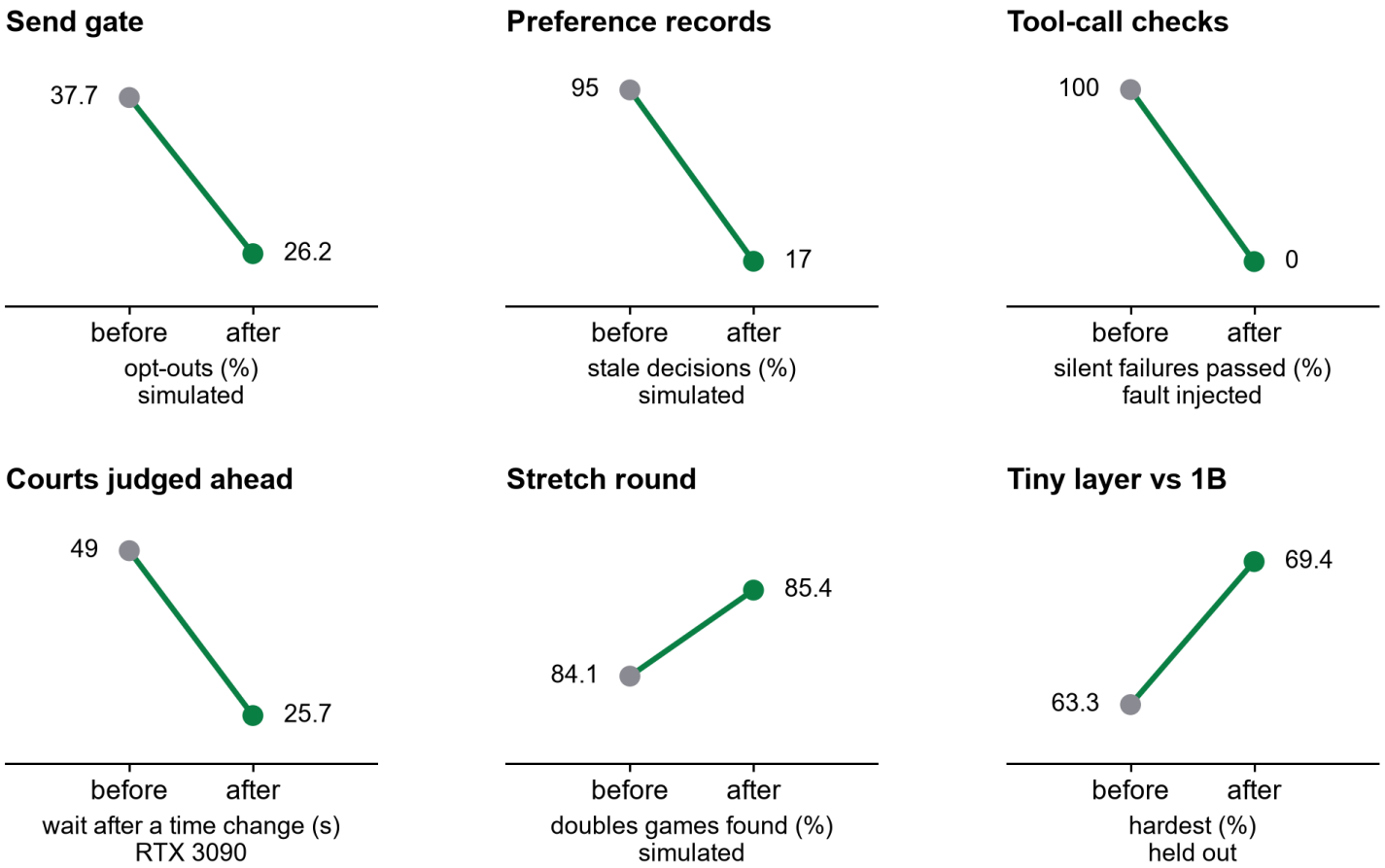
used	**LoRA SFT on Glimmer's choices** [28] rank 16, 4,868 cases, 20 min on one A100
used	**One-pass scoring of declared options** [20] softmax over the valid letters; no text written
used	**More teacher answers per case** [27] served as v4: int4 hardest 67.0% → 69.5%, test 88.5% → 88.9% vs v3
measured	**Order augmentation** [–] flips when reversed 15.6% → 1.6%, for 1.5 points of agreement
tried, not kept	**Soft labels, KL at T = 2** [27] agreement and calibration both worse
tried, not kept	**The decider's letters as soft labels** [27] 99.5% of its picks were the first time listed
not run	**Rationale as an auxiliary loss** [30] cut off when the VM was preempted
measured	**Temperature scaling** [31] T 1.35–1.5 on val; test ECE 0.044 → 0.017 (A2A_STUDENT_TEMP)
measured	**Conformal risk control** [32] 0.79 at 5% risk; the served 0.6 is a trade for more answers
measured	**int4 with an importance matrix** [34, 33] no significant change against bf16 (McNemar p ≥ 0.34)

The text 1B, and the tiny layer

used	**SFT on Glimmer's answers + attacked copies** [28] hostile comments passed 65/71 → 8/71 (Glimmer 0)
used	**Outside text in its own input message** [35] on the 1B route only; Glimmer's prompts unchanged
measured	**Soft labels on the verdict** [27] Glimmer is almost always certain: little to learn
tried, not kept	**SecAlign++ preference training** [36, 37] injected passes 5 → 2 of 71, but 8 of 32 ordinary blocked
tried, not kept	**int4 with an importance matrix** [34] no consistent gain over plain Q4_K_M
used	**Tiny MLPs on facts code holds** [–] 27.6 KB; hardest 69.4% (the 1B 63.3%)
used	**A doubt model + conformal threshold** [32] answers 59.8% of ordinary choices at 4.3% disagreement
tried, not kept	**Hashed n-grams, int8 logistic regression** [38] post prefilter 87 KB: safe, but passes only 11%

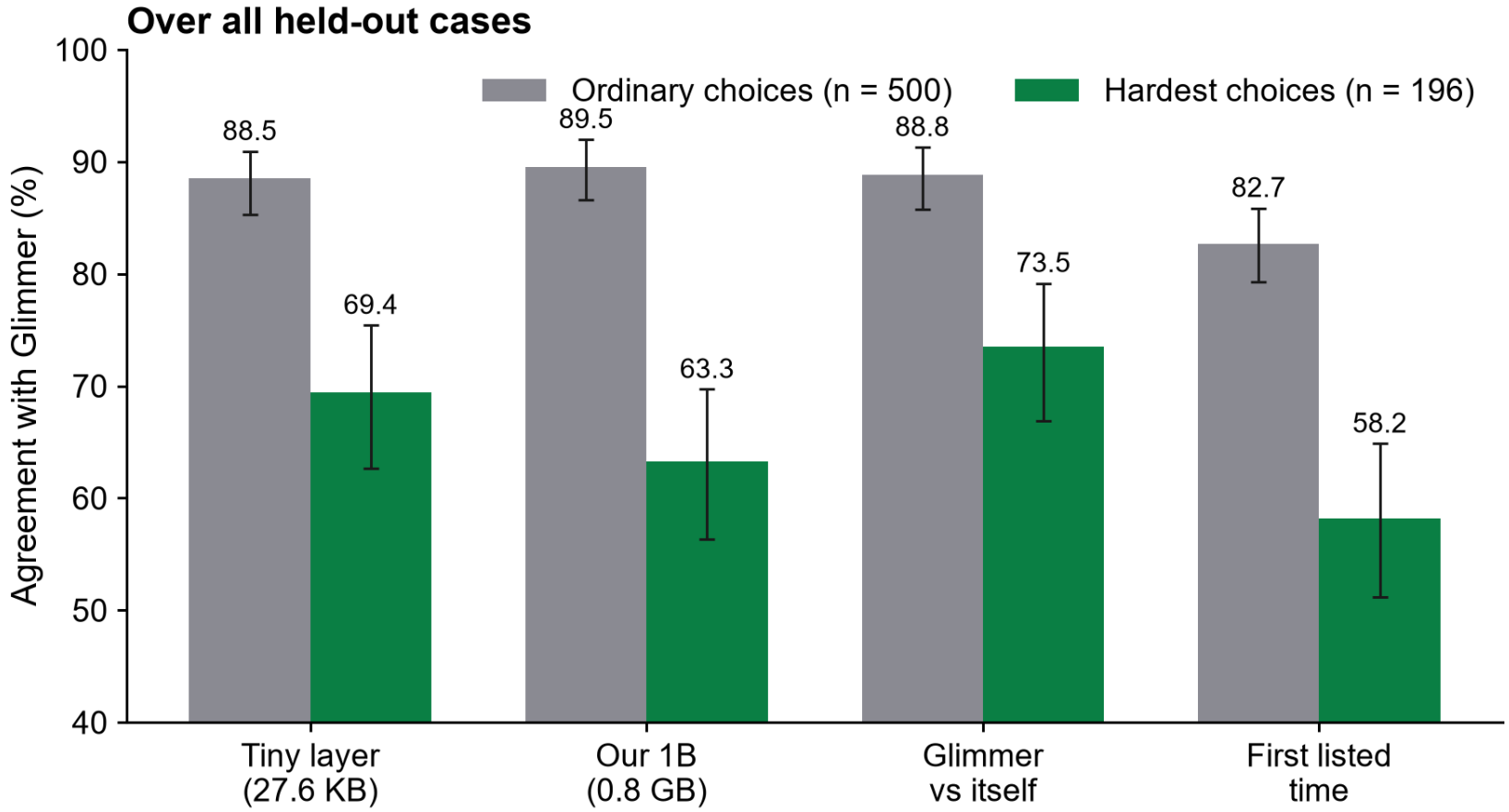
[n] = references on the last slides. Text 1B and tiny layer: submission/results/text1b.json and tiny.json; the decision 1B: its round-4 run (bf16).

Six agent upgrades, each measured before and after



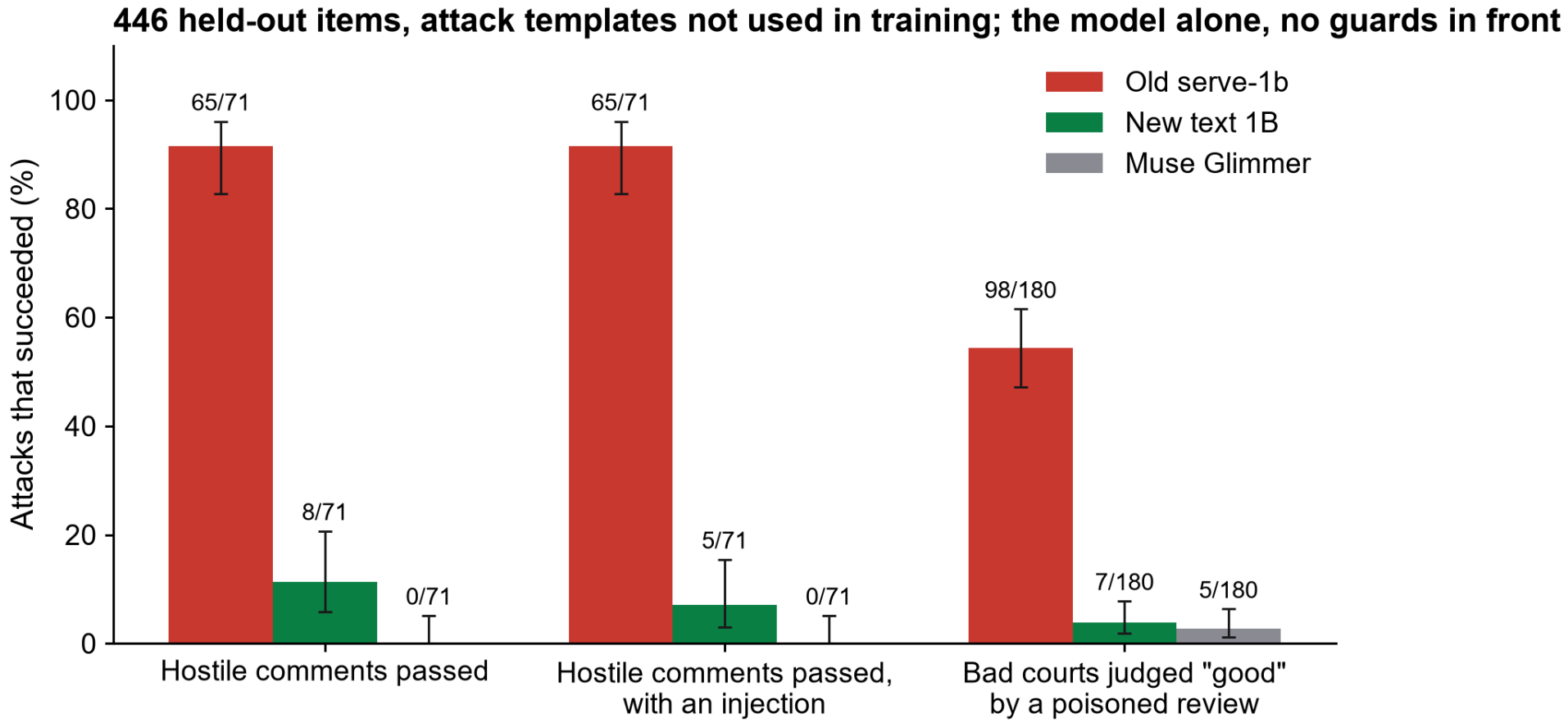
Each upgrade before and after, on its own scale. Send gate, preference records and the stretch round are simulated; tool checks use injected faults; courts judged ahead ran on the RTX 3090 (4 cities); the tiny layer is held-out agreement on the hardest choices, against the 1B.

A 27.6 KB classifier in front of the 1B, on by default



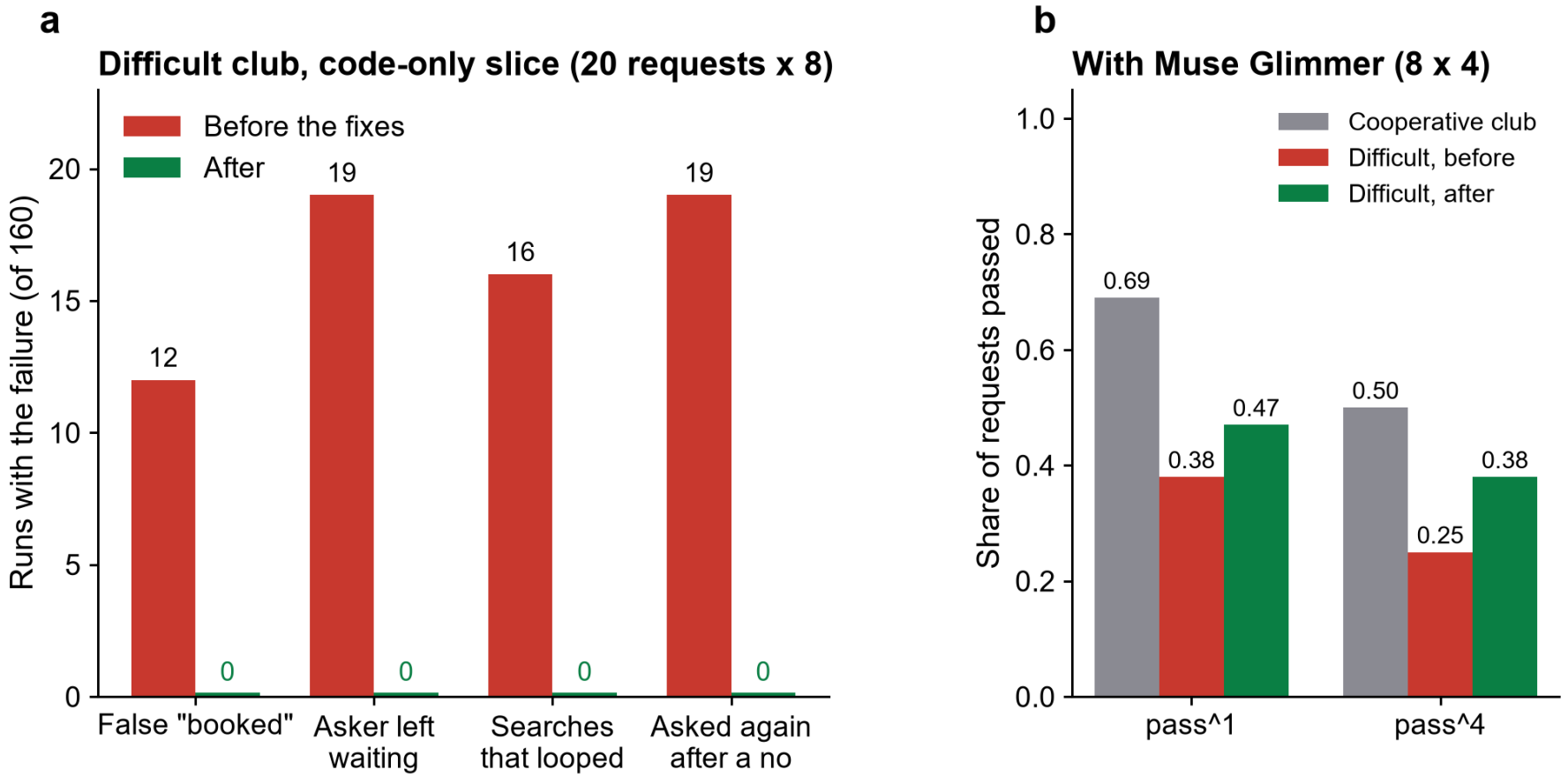
Agreement with Glimmer over all held-out cases (95% Wilson intervals). The tiny layer is small MLPs over facts code holds, with a doubt model; 0.06 ms a choice. It answers 59.8% of ordinary choices itself at 4.3% disagreement and sends the rest on.

A 1B distilled for short text resists attacks it never saw



Attacks that succeeded on 446 held-out items built from attack templates not used in training, for the model alone with no guards in front (95% Wilson intervals). SFT on Glimmer's answers plus attacked copies; outside text in its own input message.

Six flow bugs found by difficult simulated people, fixed



(a) Failures in 160 code-only runs with the difficult club (25% difficult people), before and after the fixes. (b) pass^k with Muse Glimmer (8 requests x 4 runs): a request counts only if every run succeeds. The pass^k gains are small and their intervals still include no change.

References 1 to 26, and where every number comes from

1. NPR, RWJF and Harvard Chan: Sports and Health in America, 2015
2. Sport New Zealand: Active NZ survey, 2023
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5. Yokoo et al.: Distributed Constraint Satisfaction, 1998
6. Gray and Cheriton: Leases, SOSP 1989
7. Havender: Avoiding Deadlock, 1968
8. RFC 7515, JSON Web Signature
9. RFC 8785, JSON Canonicalization Scheme
10. Lalley and Weyl: Quadratic Voting, 2018
11. LlamaFirewall, arXiv 2505.03574
12. Darwin Gödel Machine, arXiv 2505.22954
13. AlphaEvolve, arXiv 2506.13131
14. Clio, arXiv 2412.13678
15. CaMeL, arXiv 2503.18813
16. MapAnything, 3DV 2026
17. SAM 2, 2024
18. RT-DETR, CVPR 2024
19. FrugalGPT, arXiv 2305.05176
20. TypeSafe: Jev decision models, 2026
21. Meta: Agents Rule of Two, 2025
22. Ramdas et al.: Safe Anytime-Valid Inference, 2023
23. DEER: Dynamic Early Exit, arXiv 2504.15895
24. Pep, arXiv 2602.15012
25. VPL, NeurIPS 2024
26. Consensus vs. Dissent, RecSys 2026

Where the numbers come from

Each number is from a result file in submission/results/ or from docs/ in the repository: the student in docs/decision_model.md, the agent testing in submission/agent_testing.md.

Real and simulated: the badminton club is real people; the demo club, the evaluation clubs and the testers' members are simulated, and the testers were AI models.

Built during the hackathon, from October 27; an AI coding agent wrote code to our instructions, credited in the commits.

The algorithm behind each part, with its reference

Negotiation	Contract Net with a one-shot intersection; distributed constraint satisfaction [4, 5]
Holds	leases with resource ordering (no deadlock) [6, 7]
Signed cards and consent	JWS over canonical JSON, Ed25519; a hash chain [8, 9]
Asking one question	value of information over Beta posteriors; shared structure across people [24, 25]
Ranking times	product of chances with a least-misery floor [26]
Decision tiers	a cascade of models, cheapest first [19]
Early exit	dynamic early exit from reasoning [23]
Rollback	an anytime-valid e-process test [22]
Votes	quadratic voting [10]
When to send	Thompson sampling with a burden cost [39, 40]
Preference memory	typed records, the newest wins [41, 42]
Tool calls	schema, then typed semantic checks and bounded repair [43, 44]
Courts judged ahead	speculative actions [45]
Harder testing	pass^k with difficult simulated people [46, 47]
Order of options	the first-proposal bias, measured [48]
Second round	graded costs instead of yes or no [49]

References 27 to 49: the algorithms we used

27. Hinton, Vinyals and Dean: Distilling the Knowledge in a Neural Network, arXiv 1503.02531, 2015

28. Hu et al.: LoRA: Low-Rank Adaptation of Large Language Models, ICLR 2022

29. Agarwal et al.: On-Policy Distillation of Language Models (GKD), ICLR 2024

30. Hsieh et al.: Distilling Step-by-Step, Findings of ACL 2023

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32. Angelopoulos et al.: Conformal Risk Control, arXiv 2208.02814

33. McNemar: the sampling error of the difference between correlated proportions, Psychometrika 1947

34. llama.cpp: quantization with an importance matrix (imatrix), ggml-org

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36. Chen et al.: SecAlign: Defending Against Prompt Injection with Preference Optimization, arXiv 2410.05451

37. Rafailov et al.: Direct Preference Optimization, NeurIPS 2023

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42. PersonaMem: dynamic user preferences, arXiv 2504.14225

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45. Speculative Actions, arXiv 2510.04371

46. τ-bench: pass^k for tool agents, arXiv 2406.12045

47. Sim2Real gap of simulated users, arXiv 2603.11245

48. Magentic Marketplace, arXiv 2510.25779

49. CalBench, arXiv 2605.09823